# Differential Topology 1

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# Contents

# Chapter 1

# Intro

## 1.1 Topological manifolds

### Lecture 1: Syllabus day

Wed 03 Sep 2025 12:16

Smooth manifolds form a proper subset of topological manifolds. Interesting.

**Definition 1.1.1.** A topological manifold is a Hausdorff topological space $(M, \mathcal{T})$ which is second countable (it has a countable basis), and for some fixed $n \in \mathbb{N}$, for all points $p \in M$ there is an open set $p \in U \subseteq M$ and a homeomorphism $f : U \cong \hat{U} \subseteq \mathbb{R}^n$ for some $\hat{U}$. We say the dimension of $M$ is $n$, and call it an $n$-manifold.

Of course we may equivalently say for all points $p$, there is an open neighborhood $U$ and a homeomorphism $f : U \cong \mathbb{R}^n$. This follows since $\hat{U}$ admits an open cover by open balls, who have open preimages, and thus at least one has an open preimage $V$ containing $p$. Since the image of $f|V$ is an open ball, and open balls are homeomorphic to $\mathbb{R}^n$, the conclusion follows.

### Lecture 2: Smooth manifolds

Fri 05 Sep 2025 12:20

Given $f : M \to \mathbb{R}$ and a point $p$, we might want to differentiate $f$ at $p$. We can take an open neighborhood $p \in U$ and a homeomorphism $\varphi : U \cong \widetilde{U} \subseteq \mathbb{R}^n$, and try to differentiate $f \circ \varphi^{-1}$. The issue is that given another open set $V$ and another homeomorphism $\psi$, the change of variables from $f \circ \varphi^{-1}$ to $f \circ \psi^{-1}$ might not be smooth.

### Lecture 3: Smooth Manifolds

Mon 08 Sep 2025 12:22

we want to define what it means for a function $f : M \to \mathbb{R}$ to be smooth (infinitely differentiable) at some $p \in M$. Since $M$ is a manifold, there is an open set $p \in U \subseteq M$ and a homeomorphism $\varphi : U \cong \widetilde{U} \subseteq \mathbb{R}^n$, so we can consider the function $f \circ \varphi^{-1} : \mathbb{R}^n \to \mathbb{R}$. We can compute the partials of this function.

**Terminology 1.1.2.** We say $f$ is *smooth* or $C^\infty$ at $p \in U$ if $f \circ \varphi^{-1}$ is $C^\infty$ at $\hat{p} = \varphi(p) \in \widetilde{U}$.

The problem is that although $f \circ \varphi^{-1}$ might be smooth at $\hat{p}$, but given another chart $\psi : V \to \widetilde{V}$, $f \circ \psi^{-1}$ might not be smooth at $\psi(p)$. We need to add more structure to $M$ for this notion of smoothness to be well-defined.

Note that we can restrict $\psi \circ \varphi^{-1}$ to $\psi \circ \varphi^{-1} : \varphi(U \cap V) \to \psi(U \cap V)$. Of course $f \circ \psi^{-1} = f \circ \varphi^{-1} \circ \varphi \circ \psi^{-1}$. The issue can be rephrased: although $f \circ \varphi^{-1}$ might be $C^\infty$, $\varphi \circ \psi^{-1}$ might not be. The component $\varphi \circ \psi^{-1}$ is the *change of variables*, and we need the change of variables to be smooth. To ensure this is true, we *choose* a collection of homeomorphisms that are smooth, and consider that structure on the manifold, instead of taking *all* local homeomorphisms.

**Definition 1.1.3.** A *chart* for a point $p \in M$ is an open set $p \in U \subseteq M$ and a homeomorphism $\varphi : U \cong \widetilde{U} \subseteq \mathbb{R}^n$.

**Definition 1.1.4.** Let $U, V \subseteq M$ be open and $\varphi : U \to \varphi(U) \subseteq \mathbb{R}^n$, $\psi : V \to \psi(V) \subseteq \mathbb{R}^n$ be charts. We say $\varphi$ and $\psi$ are $C^\infty$-related if $\varphi \circ \psi^{-1}$ and $\psi \circ \varphi^{-1}$ are both $C^\infty$ in the sense that all of their partials exist.

**Definition 1.1.5.** Let $M$ be a topological manifold. An *atlas* $\mathcal{A}$ (sometimes a *smooth atlas*) is a collection $\{U_\alpha, \varphi_\alpha)\}_{\alpha \in A}$ of charts such that

1. The charts form an open cover of $M$:

$$M = \bigcup_{\alpha \in A} U_\alpha;$$

2. Any pair of charts is $C^\infty$.

**Terminology 1.1.6.** We often call each chart $U_\alpha, \varphi_\alpha$ a *coordinate system.*

We now know what it means for a map $f : M \to \mathbb{R}$ to be smooth: we just require it be smooth with respect to an atlas $\mathcal{A}$ for $M$. However, smooth manifolds has even more structure. One can choose *larger* atlases.

**Definition 1.1.7.** Given an atlas $\mathcal{A}$ for a manifold $M$, define the *maximal atlas* $\mathcal{A}'$ containing $\mathcal{A}$ to be the collection of *all* charts on $M$ which are $C^\infty$-related to *all* charts in $\mathcal{A}$.

This condition guarentees we always have enough charts for our atlas $M$. This leads us to

**Definition 1.1.8** (Smooth manifold)**.** A *smooth* $n$-manifold (also a $C^\infty$ $n$-manifold) is a pair $(M, \mathcal{A})$ where $M$ is an $n$-manifold and $\mathcal{A}$ is a maximal atlas on $M$.

**Example 1.1.9.** A 0-manifold is the *same* as a smooth 0-manifold and contains a countable discrete set.

Consider $1 : \mathbb{R} \to \mathbb{R}$, and consider another homeomorphism $\varphi : \mathbb{R} \to \mathbb{R}$ given by $x \mapsto x^3$. These are both certaintly charts, and $\{(\mathbb{R}, \psi)\}$ certaintly yields a smooth atlas. This is interestingly a maximal atlas??????

Consider matrices $M_{m \times n}(\mathbb{R}) \cong \mathbb{R}^{mn}$. This is also a smooth manifold.

Consider the graph of a $C^\infty$ function $f : U \subseteq \mathbb{R}^k \to \mathbb{R}^n$. For simplicity, write $\Gamma_f$ for the graph of $f$. We claim that this is a smooth manifold of dimension $k$. The natural chart is given by composing along the natural projection $\pi : \Gamma_f \to \mathbb{R}^k$. Collecting these gives an atlas.

**Example 1.1.10.** An important example is the standard smooth structure on $S^n$. Recall that we have for any $0 < j \leq n+1$, a homeomorphism $\varphi_j^\pm : U_j^\pm \to \widetilde{U}_j^\pm$ given by taking the upper hemisphere $(+)$ or lower hemisphere $(-)$ and mapping them down to a unit disk. One rigorously defines this as the product of $1_i$ for $i \neq j$, and restricting along the relevant (open) hemisphere. Alternatively, $\varphi_j^\pm : (x^1, \ldots x^n) \mapsto (x^1, \ldots x^{j-1}, x^{j+1}, \ldots x^n)$.

We claim the collection $\left\{\varphi_j^{\pm}\right\}_{j\in[n+1]\setminus[0]}$ is a smooth atlas. We check this in cases. Take $i<j$, assume WLOG that both are in the upper hemisphere, and consider $\varphi_i \circ \varphi_j^{-1}$. We show these are both smooth. To construct $\varphi_j^{-1}$, we need to re-insert the component making the norm of each vector 1. We thus have

$$\varphi_i \circ \varphi_j : (x^1, \dots, x^{j-1}, x^{j+1}, \dots, x^n) \mapsto (x^1, \dots, x^{i-1}, x^{i+1}, \dots, x^{j-1}, 1 - \sqrt{\sum_{k=1}^n (x^k)^2}, x^{j+1}, \dots, x^n).$$

This is smooth since it is a product of smooth functions. Thus, $\varphi_j$ is $C^\infty$-related to $\varphi_i$.

This argument immediately generalizes to *level sets*. Given $C^\infty$ $\Phi : U \subseteq \mathbb{R}^n \to \mathbb{R}$ with preimage $M_c \coloneqq \Phi^{-1}(c)$ as a *level set* (norm is 1) and nonzero derivatives at all $a \in M_c$, we have by the implicit function theorem that $M$ is a $C^\infty$ $(n-1)$-manifold.

**Lecture 4: awoooo**

Wed 10 Sep 2025 12:11

Let's now show $\mathbb{RP}^n$ is a topological manifold. Recall the real projective space is the quotient of $\mathbb{R}^{n+1}$ by identifying colinear $n$-tuples (in the linear algebraic sense of "colinear", meaning the line passes through 0). Chooose $U_\alpha \subseteq \mathbb{RP}^n$, and define a chart $\varphi_\alpha : U \to \mathbb{R}^n$ by scalar multiplying by $1/z^\alpha$ for $z^\alpha$ the $\alpha$th componet of a choice of representative for a point in $U_\alpha$. This is a smooth homeomorphism apparently.

The product $M \times N$ of smooth manifolds is smooth.

Let's specialize temporarily to manifolds with boundaries. We start with topological manifolds. We write $\partial M$ for the boundary of $M$. Write $\mathbb{H}^n$ for the upper half space $x \in \mathbb{R}^n$, $x^n \geq 0$.

**Definition 1.1.11.** A topological manifold with boundary is a second countable, Hausdorff space $M$ wherein every point of $M$ has a neighborhood homeomorphic to an open set of $\mathbb{H}^n$.

Manifolds with boundary then have two different types of charts: interior and boundary charts, defined in the obvious way. An *interior* point lives in some interior chart, otherwise it is a boundary point. We write $\operatorname{Int}(M)$ for the set of interior points. We need to know what it means to be smooth on a boundary chart. The solution is simply saying it is smooth at a point *if and only if* it admits an extension to an open neighborhood of a point.

## 1.2 Smooth maps

**Definition 1.2.1.** Let $M, N$ be $m$- and $n$-manifolds, respectively. A map $f : M \to N$ is *smooth* at some $p \in M$ if and only if given a chart $(U, \varphi)$ of $p$ in the atlas, and some chart $(V, \psi)$ of $f(p)$ in the atlas of $N$, the composite

$$\mathbb{R}^n \supseteq \widetilde{U} \xrightarrow{\varphi^{-1}} M \xrightarrow{f} N \xrightarrow{\psi} \widetilde{V} \subseteq \mathbb{R}^m$$

is smooth in the usual sense, *and* if $f(U) \subseteq V$. If $f$ is smooth at all $p \in M$, we say $f$ is a smooth map.

**Definition 1.2.2.** Let $M$ be a smooth manifold. A smooth map $f : M \to \mathbb{R}$ will be calleed a *smooth function on $M$*. The set of all smooth functions on $M$ is written $C^\infty(M)$.

$C^\infty(M)$ forms an $\mathbb{R}$-algebra in the usual way.

**Definition 1.2.3.** Let $f : M \to N$ be smooth. Then $f$ is a *diffeomorphism* if it is invertible and its inverses is smooth. If there is a diffeomorphism $M \to N$, we say $M$ and $N$ are *diffeomorphic*, and write $M \cong N$.

**Theorem 1.2.4.** *Composites of smooth maps are smooth.*

*Proof.* Let $f : M \to N$ and $g : N \to R$ be smooth maps. Let $p \in M$, and since $f$ is smooth, there is a chart $(U, \varphi)$ for $p$ and a chart $(V, \psi)$ of $f(p)$ with $\psi \circ f \circ \varphi^{-1}$ smooth, and $f(U) \subseteq V$. Since $g$ is smooth, there is a chart $(V', \psi')$ of $f(p)$ and a chart $(W, \xi)$ of $gf(p)$ such that $g(V') \subseteq W$ and such that $\xi \circ g \circ (\psi')^{-1}$ is smooth. If we show that $(V, \psi)$ suffices as a choice of $(V', \psi)$, then we have $gf(U) \subseteq g(V) \subseteq W$ and $\xi \circ g \circ \psi^{-1} \circ \psi \circ f \circ \varphi^{-1} = \xi \circ gf \circ \varphi^{-1}$ is smooth. Thus, it suffices to show that statement.

use the transitio map ■

**Remark 1.2.5.** doesnt matter what chart we choose cuz theyre all smoothly related!!! rigorize this

**Remark 1.2.6.** inclusions and projections (in the sense of subobjects and quotients) are differentiable

## Lecture 5: Tangent vectors

Fri 12 Sep 2025 12:21

Define $\mathbb{R}^n_a$ as the set $\{a\} \times \mathbb{R}^n$ for some vector $a \in \mathbb{R}^n$. We will show that this admits an interpretation as a tangent space to $\mathbb{R}^n$ at $a$. We say that an element $(a, v) \in \mathbb{R}^n_a$ is a *geometric tangent vector*, and we write it as $v_a$ or $v|_a$. The intuitive interpretation of this object is taking the vector $v$, and giving it starting point $a$.

The set $\mathbb{R}^n_a$ forms a vector space in the obvious way, by interpreting $\{a\}$ as 0, and simply working with the space $0 \times \mathbb{R}^n \cong \mathbb{R}^n$. The vectors $e_i|_a$ thus form a basis for this space.

This allows us to define tangent spaces to manifolds which admit embeddings into $\mathbb{R}^{n+1}$, by taking the tangent space at a point $a \in M$ as $\mathbb{R}^n_a$. However, to generalize this definition arbitrarily, we must explore some alternative representations of this information.

To do this, we must somehow translate this information into that of smooth maps and manifolds. For example, given any vector $v_a \in \mathbb{R}^n_a$, the directional derivative $D_v|_a : C^\infty(\mathbb{R}^n) \to \mathbb{R}$ given by

$$D_v|_a f = D_v f(a) = \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} f(a + tv)$$

is $\mathbb{R}$-linear, and satisfies the *product rule*

$$D_v|_a(fg) = f(a) D_v|_a g + g(a) D_v|_a f.$$

If we write $v_a = v^i e_i|_a$, then $D_v|_a f$ can be written more concretely as

$$D_v|_a f = v^i \frac{\partial f}{\partial x^i}(a).$$

This follows by the chain rule. We generalize directional derivatives using this formulation.

**Definition 1.2.7.** Let $a \in \mathbb{R}^n$. A *derivation* at $a$ is an $\mathbb{R}$-linear map $w : C^\infty(\mathbb{R}^n) \to \mathbb{R}$ satisfying the product rule

$$w(fg) = f(a)w(g) + g(a)w(f).$$

We write $T_a\mathbb{R}^n$ for the set of all derivations at $a$.

We immediately see that $T_a\mathbb{R}^n$ is a vector space by noting that if $w_1, w_2$ are derivations, then

$$(w_1+w_2)(fg) = (f(a)w_1(g)+g(a)w_1(f))+(f(a)w_2(g)+g(a)w_2(f)) = f(a)(w_1+w_2)(g)+g(a)(w_1+w_2)(f),$$

$$(cw)(fg) = c\,(f(a)w(g) + g(a)w(f)) = f(a)(cw)(g) + g(a)(cw)(f).$$

The following lemma is not too interesting, but implies a very interesting result.

**Lemma 1.2.8.** *Let $a \in \mathbb{R}^n$, $w \in T_a\mathbb{R}^n$, and $f, g \in C^\infty(\mathbb{R}^n)$. If $f$ is a constant function, then $w(f) = 0$. Additionally, if $f(a) = g(a) = 0$, then $w(fg) = 0$.*

*Proof.* For the first statement, it suffices to prove $f$ for the constant function $f(x) = 1$, since $g(x) = c$ implies $g = cf$, and the statement follows by linearity of derivations. The product rule gives

$$w(f) = w(ff) = 1w(f) + 1w(f) = 2w(f)$$

and thus $w(f) = 0$.

For the second statement, we have

$$w(fg) = f(a)w(g) + g(a)w(f) = 0w(f) + 0w(g) = 0.$$

■

**Proposition 1.2.9.** *Let $a \in \mathbb{R}^n$ and $v_a \in \mathbb{R}^n_a$. Then $D_v|_a$ is a derivation, and the map $v_a \mapsto D_v|_a$ is an isomorphism of $\mathbb{R}$-vector spaces $\mathbb{R}^n_a \cong T_a\mathbb{R}^n$.*

*Proof.* First, we show $D_v|_a$ is a derivation. This follows by the properties of the partial derivative $\partial_\mu$. To show it is $\mathbb{R}$-linear, we have

$$D_v|_a(f+g) = v^\mu \partial_\mu (f+g)|_a = v^\mu \partial_\mu f|_a + v^\mu \partial_\mu g|_a = D_v|_a f + D_v|_a g,$$

$$D_v(cf) = v^\mu \partial_\mu (cf)|_a = cv^\mu \partial_\mu f|_a = cD_v|_a f.$$

The product rule also follows by the fact that $\partial_\mu$ satisfies the product rule: $\partial_x(fg) + f\partial_x(g) + g\partial_x f$. Adding in the evaluation gives

$$D_v|_a(fg) = v^\mu \left(f(a)\partial_\mu g|_a + g(a)\partial_\mu f|_a\right) = f(a)D_v|_a g + g(a)D_v|_a f.$$

Now we must show the map $v_a \mapsto D_v|_a$ is $\mathbb{R}$-linear. To show additivity, we must show $D_{v+w}|_a = D_v|_a + D_w|_a$. This is obvious by the fact that $(v+w)^\mu = v^\mu + w^\mu$:

$$D_{v+w}|_a = (v+w)^\mu \partial_\mu|_a = v^\mu \partial_\mu|_a + w^\mu \partial_\mu|_a.$$

Similarly, since $(cv)^\mu = cv^\mu$, we have

$$D_{cv}|_a = (cv)^\mu \partial_\mu|_a = cv^\mu \partial_\mu|_a = cD_v|_a.$$

Therefore, the map is $\mathbb{R}$-linear.

The main part of the proof is that the map is an isomorphism of $\mathbb{R}$-vector spaces. Injectivity is show by showing the kernel is trivial. Since we do not know that $T_a\mathbb{R}^n$ is finite dimensional, this does not suffice as an isomorphism proof. Let $D_v|_a$ be the zero derivation $f \mapsto 0$. We want to show $v_a = 0$. Let $f$ be the coordinate function $x^j : \mathbb{R}^n \to \mathbb{R}$ by projecting onto the $j$th component. Then $\partial_i x^j = \delta_i^j$, giving

$$0 = D_v|_a = v^\mu \partial_\mu x^j|_a = v^j,$$

since $1|_a = 1$. The statement follows by letting $j$ range over $[n]$, showing the map is injective.

To prove surjectivity, let $w \in T_a\mathbb{R}^n$ be an arbitrary derivation. With the same definition of $x^j$, write $v^i = w(x^i)$. We show $w = D_v|_a$.

We show this by proving they act on maps $f : C^\infty(\mathbb{R}^n)$ in the same way. Since $f$ is analytic, it admits a Taylor expansion

$$f = f(a) + \sum_{i=1}^{n} \frac{\partial f}{\partial x^i}(a)\left(x^i - a^i\right) + \sum_{i,j=1}^{n} \left(x^i - a^i\right)\left(x^j - a^j\right) \int_0^1 (1-t) \frac{\partial^2 f}{\partial x^i \partial x^j}(a + t(x-a))\, \mathrm{d}t.$$

The functions $x^i - a^i$ and $x^j - a^j$ vanish at $x = a$. Since $w$ is $\mathbb{R}$-linear and distributes over addition, by lemma 1.2.8,

$$w((x^i - a^i)(x^j - a^j)) = 0.$$

The derivation thus annihilates the last term:

$$w(f) = w(f)(a) + \sum_{i=1}^{n} w\left(\frac{\partial f}{\partial x^i}(a)\left(x^i - a^i\right)\right) = \sum_{i=1}^{n} \frac{\partial f}{\partial x^i}(a)\left(w(x^i) - w(a^i)\right) = \sum_{i=1}^{n} \frac{\partial f}{\partial x^i}(a) w(x^i).$$

Since $w(x^i) = v^i$, this is simply $v^\mu \partial_\mu f|_a = D_v|_a$. This proves surjectivity. ■

**Corollary 1.2.10.** *$T_a\mathbb{R}^n$ has dimension $n$, and has a basis given by*

$$\left.\frac{\partial}{\partial x^i}\right|_a := D_{e_i}|_a$$

*In particular, these are defined by*

$$\left.\frac{\partial}{\partial x^i}\right|_a f = \frac{\partial f}{\partial x^i}(a).$$

*Proof.* These follow immediately by the above proof. ■

We now see an obvious way to generalize this definition to arbitrary manifolds. We can define derivations on arbitary $C^\infty$-manifolds, and then show that this is generated by the partials at a point.

**Definition 1.2.11.** Let $M$ be a smooth manifold, and let $p \in M$. A linear map $v : C^\infty(M) \to \mathbb{R}$ is a *derivation at $p$* if it satisfies the product rule

$$\forall f, g \in C^\infty(\mathbb{R}) : v(fg) = f(p)v(g) = g(p)v(f).$$

The set of all derivations for $p$ is denoted by $T_pM$, and is called the *tangent space* to $M$ at $p$. An element of $T_pM$ is a *tangent vector* at $p$.

We now prove some facts to show that this definition behaves as expected.

**Lemma 1.2.12.** *Let $M$ be a smooth manifold with or without boundary, $p \in M$, $v \in T_pM$, and $f, g \in C^\infty(M)$. If $f$ is constant, then $v(f) = 0$. If $f(p) = g(p) = 0$, then $v(fg) = 0$.*

*Proof.* The same proof for lemma 1.2.8 suffices. ■

## Lecture 6: Tangent space

Mon 15 Sep 2025 12:22

But how do we epress some $v \in T_pM$ in terms of th coordinates $x^i$ of a given chart $\varphi$ ($\varphi_i = x^i$)? Given the diagramm

$$\begin{tikzcd} U \arrow[r, "\varphi"] \arrow[d, hook] & \widetilde{U} \arrow[d, hook] \\ M & \mathbb{R}^n \end{tikzcd}$$

the tangent space induces a diagram

$$\begin{tikzcd} T_pU \arrow[r, "\varphi_*"] \arrow[d, "i_*"'] & T_p\widetilde{U} \arrow[d, "i_*"] \\ T_pM & T_p\mathbb{R}^n \end{tikzcd}$$

We would like to show there is an isomorphism $T_pM \cong T_p\mathbb{R}^n$, rendering the diagram commutative.

**Definition 1.2.13.** Let $M$ be a space and $f : M \to \mathbb{R}^n$ (often $\mathbb{R}$). The *support* of $f$, denoted $\operatorname{supp} f$, is the closure of the set of points where $f$ is nonzero:

$$\operatorname{supp} f = \overline{\{p \in M \mid f(p) \neq 0\}}.$$

If $\operatorname{supp} f \subseteq U$ for some open $U$, we say $f$ is *supported in* $U$.

**Definition 1.2.14.** Let $M$ be a smooth manifold, and $A \subseteq M$ a closed subset contained in some open set $A \subseteq U$. A continuous function $\psi : M \to \mathbb{R}$ is a *bump function* for $A$ supported in $U$ if and only if $0 \leq \psi \leq 1$, and $\psi | A = 1$ with $\operatorname{supp} \psi \subseteq U$.

**Proposition 1.2.15.** *Let $p \in M \in \mathrm{Man}$ and $v \in T_pM$. If $f, g \in C^\infty(M)$ agree on some neighborhood of $p$, then $v(f) = v(g)$. We say that $T_pM$ depends only on a neighborhood of $p$.*

*Proof.* Let $f|U = g|U$ with $p \in U$ open in $M$. Then defining $h = f - g$, we see that $h|U = 0$. Let $\operatorname{supp} h$ be some closed subset containing $U$. We claim there exists a smooth bump function for $\operatorname{supp} h$ supported on $U$. This follows by prop 2.2.5 (READ).

Let $\psi \in C^\infty(M)$ be a smooth bump function for $A = \operatorname{supp} h$ and is supported in $U$. Then $\psi h = h$ since $\psi = 1$ where $h$ is nonzero. Then $v(\psi h) = 0$ by product rule. ????????/ ■

**Proposition 1.2.16.** *Let $M$ be a smooth manifold, $U \subseteq M$ an open submanifold, $\iota : U \hookrightarrow M$ the natural inclusion, and $p \in U$. Then $\iota_* : T_pU \to T_pM$ is an isomorphism of vector spaces.*

*Proof.* Injectivity: The key is the *extension lemma*. Let $p \in V \subseteq U$ such that $\overline{V} \subseteq U$. Then jesus christ i am just going to read the book. ■

**Corollary 1.2.17.** *If $M$ is a smooth $n$-manifold, then* $\dim_{\mathbb{R}} T_pM = n$ *for any $p \in M$.*

This corollary gives a basis for the space. Let $\varphi : T_pM \cong T_{\hat{p}}\mathbb{R}^n$. Since the latter has as a basis the set $\partial_\mu|_{\hat{p}}$, meaning we have a basis in $T_pM$ by

$$\varphi_*^{-1}\left( \left.\frac{\partial}{\partial x^i}\right|_{\varphi(p)} \right).$$

This is a lot of notation, so we simply write

$$\left.\frac{\partial}{\partial x^i}\right|_p \qquad \text{or} \qquad \partial_i|_p.$$

A vector $v \in T_pM$ is then written as

$$V = V^i\partial_i|_p = V^i\varphi_*^{-1}\left(\left.\frac{\partial}{\partial x^i}\right|_p\right).$$

Note that $V^i$ are literal real numbers since they are just coefficients for the basis.

Now colnsider a manifold $M$ with boundary. The definition of $T_pM$ still works. Since the space has boundary, it is locally homeomorphic to an open set in the half space $\mathbb{H}^n$. This includes into $\mathbb{R}^n$, and since smooth functions on the half space must extend to smooth maps on $\mathbb{R}^n$, we still get our definition.

Here's a useful special case. Let $V$ a vector space and $a \in V$. There exists a natural isomorphism of vector spaces $V \to T_aV$ given by $a \mapsto D_v|_a$ natural in both $V$ and $a$. This map is natural $1_{\mathbb{R}\text{-}\mathcal{V}\mathrm{ect}} \cong T_{(-)}(-)$.

Does the tangent space functor preserve products?

**Proposition 1.2.18.** *$T$ preserves products (3.14).*

*Proof.* I think what we do is bring it into $\mathbb{R}^n$, distribute there, and pull back.23212 ■

## Lecture 7: Tangent bundles

Fri 19 Sep 2025 12:20

I MISSED STUFF. WE PICKED UP WITH CHANGE OF COORDINATES IN TANGENT SPACES $T_pM$.

Choose $p \in M$, and choose a chart $(U, \varphi = u^\mu)$ for $p$. This is a coordinate system, and $\varphi(p) = \hat{p} = (x^1, \ldots, x^n)$ in the natural basis $\partial_\mu|_{\hat{p}}$. This gives a vector

$$\left.\frac{\partial}{\partial x^i}\right|_p = \varphi^{-1}\left(\left.\frac{\partial}{\partial x^i}\right|_{\hat{p}}\right) \implies v = v^\mu\partial_\mu.$$

However, given any other chart $(V, \psi)$, we get another basis induced in the same way. To change variables, we use the precomposition-esk change of variables map $(\psi \circ \varphi^{-1})_*$, which is literally the change of variables in the calc 3 sense.

This change of variables basically becomes the chain rule.

**Lemma 1.2.19.** *Every $v \in T_pM$ is tangent to some smooth curve in $M$.*

*Proof.* Let $(p \in U, \varphi)$ a chart of the curve at $p$. Then

$$v = v^i \left.\frac{\partial}{\partial x^i}\right|_p.$$

Let $\gamma : (-\varepsilon, \varepsilon) \to M$ have $\gamma_i(t) = tv^i$. Then $\gamma(0) = p$ and $\gamma$ is $C^\infty$. Add idk i think we are writing $\gamma'$ instead which is $\gamma_*\frac{\mathrm{d}}{\mathrm{d}t}$ eval at $p$? ■

**Definition 1.2.20.** Let $M$ be a smooth manifold. Define the *tangent bundle* $TM$ as the disjoint union

$$TM \coloneqq \coprod_{p \in M} T_pM.$$

The motivation here is smooth vector fields. There is a projection $TM \to M$ by mapping each $v \in T_pM \subseteq TM$ to $p$:

$$\pi : T_pM \mapsto p.$$

If we give $TM$ a smooth structure, we can define some $V : M \to TM$ to be a smooth vector field if $\pi \circ V = 1$.

**Proposition 1.2.21.** *For any smooth $n$-manifold $M$, the tangent bundle has a natural smooth structure making it a $2n$-manifold, and the canonical projection $\pi : TM \to M$ is smooth.*

### Lecture 8: Tangent bundles

Mon 22 Sep 2025 12:20

We want to give $TM$ a smooth manifold structure such that proposition 1.2.21 is true. Our idea is if $(U, \varphi)$ is a chart, consider $\pi^{-1}(U)$, which is the set of tangent vectors at $p \in U$ for all $p$. We can define a function

$$\widetilde{\varphi} : \pi^{-1}(U) \to \widetilde{U} \times \mathbb{R}^n \subseteq \mathbb{R}^{2n}, \qquad T_pM \ni v \mapsto (\varphi(p), v^1, \ldots, v^n),$$

where $v^i$ is given under the chart $(U, \varphi)$. We want to topologize $TM$ using the maps $\widetilde{\varphi}$. To do this, we demand that $\widetilde{\varphi}$ be a homeomorphism. We recall a lemma from earlier in the book.

**Lemma 1.2.22.** *Let $M$ be a set, and $\{U_\alpha\}_{\alpha \in A}$ a collection of subsets of $M$, together with injective maps $\varphi_\alpha : U_\alpha \to \mathbb{R}^n$ for all $\alpha \in A$, such that the following holds.*

1. *For all $\alpha \in A$, $\varphi_\alpha(U_\alpha)$ is open.*
2. *For all $\alpha, \beta \in A$, $\varphi_\alpha(U_\alpha \cap U_\beta)$ and $\varphi_\beta(U_\alpha \cap U_\beta)$ is open.*
3. *Whenever the intersection is nonempty, $\varphi_\alpha \circ \varphi_\beta^{-1}$ is a diffeomorphism.*
4. *Countably many $U_\alpha$s cover $M$.*
5. *For all $p \neq q \in M$, either there is some $U_\alpha \ni p, q$, or there are disjoint sets $U_\alpha, U_\beta$ with $p \in U_\alpha$ and $q \in U_\beta$.*

*Then $M$ has a unique smooth structure such that each $(U_\alpha, \varphi_\alpha)$ is a smooth chart.*

*Proof.* Topologize by generating a topology on preimages of each $\varphi_\alpha$ on open sets. Second countability follows by (4). Hausdorff follows by (5) and the definition of open sets. Since each $\varphi_\alpha$ is injective, factoring through the image gives $\varphi_\alpha : U_\alpha \to \varphi_\alpha(U_\alpha)$ is a homeomorphism. We then obtain the smooth transition maps by (2) and (3). ∎

Recall our situation with $TM$. For each chart $(U, \varphi)$ of $M$, there is an injective map $\widetilde{\varphi} : \pi^{-1}(U) \to \widetilde{U} \times \mathbb{R}^n$. Then the collection of $(\widetilde{\varphi}, \pi^{-1}(U))$ satisfies the requirements of lemma 1.2.22. Smoothness of $\pi$ is immediate. Additionally,

**Proposition 1.2.23.** *If $M$ is a smooth $n$-manifold which can be covered by one chart, then there is a diffeomorphism $TM \cong M \times \mathbb{R}^n$.*

*Proof.* Follows by definition, since there is a chart $\widetilde{\varphi} : TM \to M \times \mathbb{R}^n$ since the chart covers the manifold. The global chart is a diffeomorphism. We then say that $TM$ is a trivial bundle. ∎

The upshot of this is for all $p \in M$, there is a neighborhood $U$ of $p$ with $\pi^{-1}(U) \cong U \times \mathbb{R}^n$. Thus, $TM$ is locally trivial.

This puts constraints on what $TM$ can be. Consider $TS^1$. If we imagine gluing a line to each point of $S^1$ perpendicular to the plane of the circle, we get a cylinder. This is correct, since $TS^1 \cong S^1 \times \mathbb{R}$. However, we could also glue them by rotating the direction as we traverse around the circle, giving us a Mobïus strip, which we cannot have since it is not locally trivial.

**Definition 1.2.24.** Let $F : M \to N$ be a smooth map. Define $F_* = \mathrm{d}F$, called the *global pushforward* or *global differential* by $\mathrm{d}F : TM \to TN$ via $\mathrm{d}F|T_pM = \mathrm{d}F_p$. Alternatively,

$$\mathrm{d}F = \coprod_{p \in M} \mathrm{d}F_p.$$

This still behaves covariantly.

**Definition 1.2.25.** A smooth vector field $V$ is a smooth map $M \to TM$ such that $\pi \circ V = 1$.

**Definition 1.2.26.** Let $F : M \to N$ be a smooth map. Then the *rank* of $F$ at $p$. is the rank of th matrix representation of $\mathrm{d}F_p : T_pM \to T_pN$. If the rank is constant with respect to $p$, we say the rank is constant, and simply refer to the *rank* of $F$. The rank at $p$ is written $\mathrm{rk}_p F$.

Clearly, given $F : M \to N$,

$$0 \leq \mathrm{rk}_p F \leq \min(\dim M, \dim N).$$

If $\mathrm{rk}_p F = \min(\dim M, \dim N)$, then we say $F$ has *full rank* at $p$.

**Definition 1.2.27.** A smooth map $F : M \to N$ is a *submersion* if and only if $F$ is constant rank, and $\mathrm{rk} F = \dim N$. We say $F$ is a *immersion* if and only if $F$ is constant rank, and $\mathrm{rk} F = \dim M$.

**Proposition 1.2.28.** *Let $F : M \to N$ be smooth, and let $p \in M$. If $\mathrm{d}F_p : T_pM \to T_{F(p)}N$ is surective, then there is a neighborhood $U$ of $p$ such that $F|U$ is a submersion. If it is injective, then $F|U$ is an immersion.*

*Proof.* Obvious, since the rank of surjections is just dim of codomain, similarly for injections, and continuity of $F$ implies there is a neighborhood wherein rank is constant. And also the fact that the space of full rank matrices is open. ∎

For example, the projection $\pi : M \times N \to M$ is surjective, so it is surjective on all tangent spaces. Thus, $\pi$ is a smooth submersion. An immersion could be given by a smooth path $\gamma : I \to M$ if and only if it has everywhere nonzero velocity, since $\mathrm{d}\gamma$ is non-rank-0 whenever velocity doesn't vanish everywhere. A fun example is $\pi : TM \to M$ is a smooth submersion. This follows since its a local property, so we apply the definition.

## Lecture 9: Submersions and immersions

Wed 24 Sep 2025 12:24

The linear map $\mathrm{d}F$ as a matrix in a basis is the Jacobian

$$J_{ij} = \frac{\partial \hat{F}^i}{\partial x^j}.$$

**Definition 1.2.29.** Let $F : M \to N$ be smooth. Then $F$ is a *local diffeomorphism* if and only if for all $p \in M$, there is some $p \in U \subseteq M$ open such that $F(U)$ is open in $N$, and $F|U$ is a diffeomorphism onto its image.

For example, consider the covering $\mathbb{R} \to S^1$. This is clearly not a diffeomorphism, but it is a local diffeomorphism since we just restrict to a small enough neighborhood such that it becomes injective.

**Theorem 1.2.30.** *Let $M, N$ be smooth manifolds, $F : M \to N$ smooth, and $p \in M$ such that $\mathrm{d}F_p$ is invertible. Then there is a connected neighborhood $U_0$ of $p$, and $V_0$ of $F(p)$, such that $F|U_0 : U_0 \cong V_0$ is a diffeomorphism.*

Here are some obvious properties of local diffeomorphisms.

- Composites of local diffeomorphisms are locally diffeomorphic.
- Finite products of local diffeomorphisms are locally diffeomorphic.
- Diffeomorphisms are local diffeomorphisms.
- Every bijective local diffeomorphism is a diffeomorphism.

**Proposition 1.2.31.** *Let $F : M \to N$ be smooth. Then $F$ is a local diffeomorphism* if and only if *$F$ is an immersion and a submersion. Moreover, if $\dim M = \dim N$ and $F$ is either an immersion or a submersion, then it is a local diffeomorphism.*

**Theorem 1.2.32** (Rank)**.** *Let $F : M^m \to N^n$ be smooth and constant rank $\operatorname{rk} F = r$. Then for all $p \in M$, there are two smooth charts $(U, \varphi)$ centered at $p$ (meaning $\varphi(p) = 0$), and $(V, \psi)$ or $N$ centered at $F(p)$, such that $(\psi \circ F \circ \varphi^{-1})(x^1, \ldots, x^m) = (x^1, \ldots, x^r, 0, \ldots, 0)$.*

*In particular, if $F$ is a submersion ($n \leq m$), then $\hat{F}(x^1, \ldots, x^m) = (x^1, \ldots, x^n)$. If $F$ is an immersion ($m \leq n$), then $\hat{F}(x^1, \ldots, x^m) = (x^1, \ldots, x^n)$. Smooth submersions thus look like projections, and smooth immersions look like injections. Constant rank maps are thus in particular composites of submersions and immersions.*

*Proof.* WLOG, take $M = U \subseteq \mathbb{R}^n$ open, $V \subseteq \mathbb{R}^n$ open, and let $F : U \to V$ be smooth. ■

### Lecture 10: The rank theorem

Fri 26 Sep 2025 12:21

The rank theorem tells us that immersions are related to embeddings, and submersions are related to quotients. Indeed, we *define* smooth embeddings as embeddings which are also immersions. Of course, these are just injective immersions.

### Lecture 11: ??

Mon 29 Sep 2025 12:36

**Theorem 1.2.33.** *Let $F : M \to N$ smooth. Then $F$ is a smooth immersion if and only if every $p \in M$ has a neighborhood $U \subseteq M$ such that $F|U$ is a smooth embedding.*

**Theorem 1.2.34** (Local section theorem)**.** *Let $\pi : M \to N$ be smooth. Then $\pi$ is a smooth submersion if and only if every $p \in M$ is in the image of a smooth local section of $\pi$.*

Smooth submersions are open, and if they are surjective then they are quotients.

**Lecture 12: Smooth coverings, embedded manifolds**

Wed 01 Oct 2025 12:21

**Theorem 1.2.35.** *Let $\pi : M \to N$ be a smooth submersion. Then any $F : N \to P$ is smooth if and only if $F \circ \pi$ is smooth. This is preisely universality of quotients in* $\mathcal{M}\mathrm{an}$.

**Theorem 1.2.36.** *Let $\pi : M \to N$ be a smooth submersion. If $F : M \to P$ is smooth and constant on each fiber $\pi^{-1}(n)$ of $\pi$, then there is exactly one $\widetilde{F} : N \to P$ such that*

$$\begin{tikzcd} M \arrow[r, "F"] \arrow[d, "\pi"'] & P \\ N \arrow[ur, dashed, "\widetilde{F}"'] & \end{tikzcd}$$

*commutes.*

*Proof.* Exists by topology, smooth by theorem 1.2.36. ∎

**Theorem 1.2.37.** *Let there be a diagram of solid arrows*

$$\begin{tikzcd} & M \arrow[dl, "\pi_1"'] \arrow[dr, "\pi_2"] & \\ N_1 \arrow[rr, dashed, "F"'] & & N_2 \end{tikzcd}$$

*where each solid arrow is a submersion which are constant on each others fibers. Then there is a unique diffeomorphism as indicated by the dashed arrow, rendering the diagram commutative.*

**Definition 1.2.38.** Recall that a covering space of a space $M$ is a surjective, continuous map $\pi : E \to M$ between connected and locally path-connected spaces, such that for all $p \in M$, there is a neighborhood $U$ of $p$ such that each component of $\pi^{-1}(U)$ is mapped homeomorphically to $U$. We say $U$ is *evenly covered*. If $E$ is also simply connected (all loops contract to a point), we say it is a *universal cover*.

**Proposition 1.2.39.** *Let $\pi : E \to M$ be a smooth covering map. Given any evenly covered open $U \subseteq M$, any $q \in U$, and any $p \in \pi^{-1}(q)$, then there is a* unique *local section $\sigma : U \to E$ such that $\sigma(q) = p$.*

**Proposition 1.2.40.** *Let $M$ be a smooth manifold and $\pi : E \to M$ a (not necessarily smooth) covering. Then $E$ is a topological $n$-manifold with a unique smooth structure promoting $\pi$ to a smooth covering.*

**Corollary 1.2.41.** *If $M$ is a connected smooth manifold, then there is a simply connected smooth manifold $\widetilde{M}$ together with a smooth map $\pi : \widetilde{M} \to M$. We call $\widetilde{M}$ the universal smooth cover.*

# Chapter 2

# Submanifolds

## 2.1 Embedded submanifolds

**Definition 2.1.1.** Let $M$ be a smooth manifold, and $S \subseteq U$ a subspace endowed with a smooth structure such that the inclusion $S \hookrightarrow U$ is a smooth embedding. Then $S$ is an *embedded submanifold*.

### Lecture 13: Submanifolds

Fri 03 Oct 2025 12:20

For example, $S^n \hookrightarrow \mathbb{R}^n$ is an embedded submanifold. The main proposition is

**Proposition 2.1.2.** *Let $F : N \to M$ be a smooth embedding. Then $\operatorname{im} F$ is an embedded submanifold of $M$ with a unique smooth structure.*

*Proof.* Endow $\operatorname{im} F$ with the smooth structure on $N$ postcomposed by $F$ or something. ■

Let $U \subseteq M$ and $f : U \to M$ smooth. Then $(1, f) : U \to M \times N$ is a homeomorphism onto the image $f \subseteq M \times N$. This is a section of the projection $\pi : M \times N \to M$, so $(1, f)$ is an immersion. We then have that the graph is an embedded submanifold. We will see that embedded submanifolds look locally like graphs of functions later.

**Definition 2.1.3.** An embedded submanifold $S \subseteq M$ is *properly embedded* if and only if $S \hookrightarrow M$ is a proper map (preimage of compact sets is compact).

**Proposition 2.1.4.** *Let $S \hookrightarrow M$ be an embedded submanifold. Then $S$ is a proper embedding if and only if $S$ is closed in $M$.*

*Proof.* If $S \hookrightarrow M$ is properly embedded, then the inclusion is closed since manifolds are locally compact Hausdorff (Lee A.57). Conversely, Lee A.53. ■

In particular, every compact/closed embedded submanifold is a proper embedding. In particular,

**Proposition 2.1.5.** *Let $F : M \to N$ be smooth. Then the graph is properly embedded in $M \times N$.*

Proven in the homework last year that graphs of continuous maps are closed.

**Definition 2.1.6.** A $k$-dimensional slice $S$ of $U \subseteq \mathbb{R}^n$ is of the form $\{(x^\mu, c^\nu)\}$ for the $c^\mu$s constant.

**Definition 2.1.7.** Let $M$ be a smooth manifold with a chart $(U, \varphi)$. If $S \subseteq U$ such that $\varphi(S)$ is a $k$-slice of $\varphi(U)$, then $S$ is a $k$-slice of $U$.

**Theorem 2.1.8** (Slice charts)**.** *All embedded submanifolds locally look like $\mathbb{R}^k \hookrightarrow \mathbb{R}^{n \geq k}$. In particular, if $S \hookrightarrow U$, then for all $p \in S$ there is a chart $(U, \varphi)$ such that $S \cap U$ is a $k$-slice of $U$.*

proof is like rank thm looks like slices.

**Definition 2.1.9.** Let $\Phi : M \to N$ be smooth. Then for any $c \in N$, the fiber $\Phi^{-1}(c)$ is the *level set of $\Phi$ at $c$.*

**Proposition 2.1.10.** *Let $\Phi : M \to N$ be smooth with constant rank. Then the level sets of $\Phi$ are properly embedded submanifolds in $M$.*

### Lecture 14: More submanifolds

Mon 06 Oct 2025 12:21

Recall the main theorem. The slice charts theorem tells us that an embedded submanifold has nice charts which look like $(x^1, \ldots, x^k) \mapsto (x^1, \ldots, x^k, 0, \ldots, 0)$.

**Definition 2.1.11.** Let $p \in M$ and $\Phi : M \to N$ smooth. $p$ is a *regular point* of $\Phi$ if $\mathrm{d}_p\Phi : T_pM \to T_{\Phi(p)}N$ is surjective. Otherwise, $p$ is a *critical point* of $\Phi$. A $c \in N$ is a *regular value* of $\Phi$ if $p$ is a regular point for all $p \in \Phi^{-1}(c)$. If $c$ is a regular value of $\Phi$, then the fiber of $c$ is a *regular level set* of $\Phi$.

**Corollary 2.1.12.** *Every regular level set between smooth manifolds is a proper embedded submanifold of codimension equal to the dimension of the domain.*

### Lecture 15: Lie groups

Wed 08 Oct 2025 12:20

We conclude submanifolds by discussing their tangent spaces. Let $S \hookrightarrow M$ be an immersed submanifold. The inclusion $i$ raises to an injection $\mathrm{d}_p(i) : T_pS \to T_pM$ for each $p \in S$. Since its injective, we may identify it with its image up to isomorphism. This idea yields a useful proposition.

**Proposition 2.1.13.** *Let $p \in S \hookrightarrow M$ be an embedded submanifold. Then $T_pS$ is isomorphic to the subset $K$ of $T_pM$ such that for each $v \in k$ and $f \in C^\infty(M)$ which vanishes on $S$ ($f|S = 0$), we have $v(f) = 0$.*

*Proof.* Let $v \in T_pS$ identified with a subspace of $T_pM$. Then there is exactly one $w \in T_pS$ with $v = \mathrm{d}_p(i)(w)$. If $f|S = 0$, then $v(f) = w(f \circ i) = w(0)$. For the converse, let $v \in T_pM$ satisfying the criteria. Choose slice coordinates $(x^1, \ldots, x^n)$ for $S$ in a neighborhood $U$ of $p$. Then $i : S \cap U \to M$ has coordinate representation $\hat{i}(x^1, \ldots, x^k) = (x^1, \ldots, x^k, 0, \ldots, 0)$. Then

$$v = v^\mu \partial_\mu|_p.$$

We have $v^j = 0$ whenever $j > k$. Choose some $f(x)$ with $f|U \cap S = 0$. It just kind of follows. ∎

**Corollary 2.1.14.** *Let $S \hookrightarrow M$ be an embedded submanifold, $U \subseteq M$, and $\Phi : U \to N$ any local defining map for $S$ (meaning $\Phi^{-1}(c) = S \cap U$ for some $c$). Then $T_pS = \operatorname{Ker} \mathrm{d}_p\Phi$.*

This is quite useful. Suppose that $S \subseteq M$ is a level set of some smooth submersion $\Phi = \Phi^\mu : M \to \mathbb{R}^k$. Then any $v \in T_pM$ is also tangent to $s$ if $v(\Phi^i) = 0$ for all $i$.

# Chapter 3

# Lie groups

**Definition 3.0.1.** A *Lie group* is a topological group $G$ with a smooth structure such that multiplication and inversion are smooth. Equivalently, $G$ is a group object in the categorty of smooth manifolds.

**Proposition 3.0.2.** *Let $G$ be a smooth manifold together with a smooth map $G \times G \to G$ given by $(g, h) \mapsto gh^{-1}$. Then $G$ is a Lie group, and mulitplication and inversion are recovered in the obvious way.*

**Definition 3.0.3.** A *left translation* by $g \in G$ is the map $L_g : G \to G$ given by $h \mapsto gh$. Right translations similarly are $R_g : h \mapsto hg$. They are diffeomorphisms since $L_g^{-1} = L_{g^{-1}}$.

A useful example is $\mathrm{GL}(n, \mathbb{R})$ or $\mathrm{Gl}(n, \mathbb{C})$. These are of dimension $n^2$ and $2n^2$, respectively. Most useful Lie groups arise as subgroups of these. We can also consider $\mathbb{R}^n, \mathbb{C}^n$ under addition or $\mathbb{R}^*, \mathbb{C}^*$ under multiplication. The latter have Lie subgroups $R_{>0}$ and $S^1$.

**Proposition 3.0.4.** *Let $G, H$ be Lie groups. Then $G \times H$ is a Lie group.*

**Definition 3.0.5.** A *discrete* Lie group is a 0-dimensional Lie group.

**Definition 3.0.6.** Lie groups form a category $\mathcal{L}ie$. A morphism $F : G \to H$ of Lie groups is a smooth group homomorphism.

Exponentiation $(x^1, \ldots, x^n) \mapsto (e^{x^1}, \ldots, e^{x^n})$ is a Lie group homomorphism $\mathbb{R}^n$ to the torus $\mathbb{T}^n$. Conjugation is smooth and is thus a Lie group homomorphism.

**Theorem 3.0.7.** *Lie group homomorphisms have constant rank.*

*Proof.* Choose $g \in G$ a Lie group with identity 1. Since $F : G \to H$ is a Lie group hom, $F(1) = 1$. We will show the rank of $\mathrm{d}_g F$ is $\mathrm{d}_1 F$, and this suffices. Choose $h \in G$, and consider $(F \circ L_g)(h) = F(gh) = (L_{F(g)} \circ F)(h)$. Thus,

$$F \circ L_g = L_{F(g)} \circ F.$$

We can pass to the differential

$$\mathrm{d}_1(F) \circ \mathrm{d}_1 L_g = \mathrm{d}_1(L_{F(g)}) \circ \mathrm{d}_1(F).$$

Since each $L$ is a diffeomorphism, we see that the differentials of $L$ are constant rank. Apparently it just follows idk check the book. ■

**Corollary 3.0.8.** *A Lie group homomorphism is an isomorphism if and only if it is bijective.*

This follows by the global rank theorem.

**Remark 3.0.9.** The kernel $\operatorname{Ker} F$ of a Lie group homomorphism is the *level set* $F^{-1}(1)$. The fact that $F$ is constant rank tells us that the kernel is a *properly embedded submanifold* of the domain. Since the kernel is also a group, it should be a Lie group.

## Lecture 16: More Lie groups

Fri 10 Oct 2025 12:22

**Definition 3.0.10.** A Lie subgroup is a subgroup with a smooth structure and topology such that it is a Lie group, and inclusion is an immersed submanifold.

**Proposition 3.0.11.** *Let $G$ be a Lie group and $H$ a subgroup that is an* embedded *submanifold.*

*Proof.* It suffices to show $H$ is a Lie group. This follows easily by a diagram chase. ∎

**Lemma 3.0.12.** *Let $G$ be a Lie group and $H$ an* open *subgroup. Then $H$ is an embedded Lie group, and $H$ is closed, and $H$ is a union of connected components.*

*Proof.* We get that $H$ is an embedded submanifold of codimension 0, and we appeal to proposition 3.0.11. The coset $gH$ is the image of left multiplication $Lg$, which is a diffeomorphism, so the image of the open set $H$ is open. We of course have $G \setminus H$ is the union of all nontrivial cosets, so $H$ is the compliment of the open set $G \setminus H$. The final follows from usual topology: any clopen set is a union of path components. Of course this coencides with path connectedness in manifolds. ∎

For example, given an open subset $U$ of $S^1$ containing 1, the group $\langle U \rangle$ is an open subgroup of $G$.

**Proposition 3.0.13.** *Let $G$ be a Lie group, and $W \subseteq G$ a neighborhood of 1. Then $W$ generates an open subgroup of $G$. If $W$ is connected, then $W$ generates a connected open subset of $G$. If $G$ is connected, then $W$ generates $G$.*

*Proof.* Let $H = \langle W \rangle$. Define

$$W_k = \left\{ g \in G \mid \exists \{w_i\}_{i=1}^{\ell} \subseteq W, \ell \leq k \text{ s.t. } g = \prod_{i=1}^{\ell} w_i \right\}.$$

Then

$$H = \bigcup_{k \geq 1} W_k.$$

Since inversion is a diffeomorphism, $W^{-1} = i(W)$ is open. Of course $W_1 = W \cup W^{-1}$, and $W_k = W_{k-1}W_1$. This is just

$$W_k = W_{k-1}W_1 = \bigcup_{g \in W_1} L_g(W_{k-1}).$$

Since $L_g$ is a diffeomorphism, each of these sets is open, so $W_k$ is open. Subgroup follows by $1 \in W$. $H$ is then a Lie subgroup by lemma 3.0.12. If $W$ is connected, then so is $W^{-1}$, and thus $W_1 = W \cup W^{-1}$ is connected since $1 \in W \cap W^{-1}$. We then see

$$W_2 = \mu(W_1 \times W_1)$$

is connected since $\mu$ is continuous. Inducting gives $W_k$ is connected. Since each of these sets contains 1, they are not disjoint, so $H$ is connected. If $G$ is connected, then $H$ is an open and closed subgroup so $H = G$. ∎

**Definition 3.0.14.** Let $G$ be a Lie group, and let $G_0 \subseteq G$ be the connected component containing 1. We call $G$ the *identity component of* $G$.

**Proposition 3.0.15.** *Let $G$ be a Lie group and $G_0$ the identity component. Then $G_0$ is normal in $G$, and is the* only connected subgroup. *Every connected component of $G$ is diffeomorphic to $G_0$.*

The idea of the proof is that connected components are cosets. Act by $L_g$ and it falls out.

**Proposition 3.0.16.** *If $F : G \to H$ is a Lie group homomorphism, then* $\operatorname{Ker} F$ *is a properly embedded Lie subgroup of codimension* $\operatorname{rk} F$.

*Proof.* We get subgroup from group theory. The kernel is the fiber of the identity. Since $F$ has constant rank, the kernel is a properly embedded submanifold. ∎

**Proposition 3.0.17.** *let $F : G \to H$ be an injective Lie group homomorphism. Then $F(G)$ has a unique smooth structure such that it is a Lie subgroup of $H$. Furthermore, $G \cong F(G)$.*

*Proof.* Because $F$ has constant rank, it is (easily shown to be) an injective smooth submersion. Therefore, its image admits a unique smooth structure making it an immersed submanifold, and thus a Lie subgroup. The rest follows. ∎

For example, $\mathrm{SL}(n, \mathbb{R})$ is the kernel of $\det : \mathrm{GL}(n, \mathbb{R}) \to \mathbb{R}$.

**Theorem 3.0.18.** *Let $G$ be a Lie subgruop with $H \subseteq G$ a Lie subgroup. Then $H$ is closed in $G$ if and only if it is embedded.*

It will be useful to consider *smooth* group actions of Lie groups on smooth manifolds. The best formalism of this is as a smooth map $G \times M \to M$ which induces the relevant group homomorphism.

Consider $O(n)$ as the subset of $\mathrm{GL}(n, \mathbb{R})$ such that $\langle Ax, Ay \rangle = \langle x, y \rangle$. We call this the orthogonal group. We will define this somehow eventually.

## Lecture 17: More more Lie groups

Tue 14 Oct 2025 12:20

Suppose we have $H \subseteq G$ a Lie subgroup. Then $H$ acts smoothly on $G$ by right , multiplication or left multiplication. The orbits under this orbit are cosets. We call the stabilizer of $p \in M$ the *isotropy* (sub)group at $p$.

**Definition 3.0.19.** Let $M, N$ be manifolds with $G$-actions $\theta : G \times M \to M$, $\psi : G \times N \to N$ for $G$ a Lie group. A *smooth map* $F : M \to N$ is called *$G$-equivariant* if the diagram

$$\begin{tikzcd} M \arrow[r, "F"] \arrow[d, "\theta_g"'] & N \arrow[d, "\psi_g"] \\ M \arrow[r, "F"'] & N \end{tikzcd}$$

commutes for each $g \in G$. The (functor) category $\mathcal{S}\mathrm{mth}^G$ is the category of smooth $G$-manifolds together with $G$-equivariant maps.

**Theorem 3.0.20** (Equivariant rank theorem). *Let $F : M \to N$ be a smooth $G$-equivariant map for $M, N$ $G$-manifolds, $G$ a Lie group. If the action on $N$ is transitive, then $F$ is constant rank, the action on $M$ is transitive, and if $F$ is (sur/in/bi)jective then it is (sub/im)mersion/diffeomorphism.*

**Proposition 3.0.21.** *Given a $G$-action $\theta$ on $M$, for all $p \in M$ the orbit map $\theta^{(p)} : G \to M$ given by $g \mapsto gp$ is smooth and has constant rank. Since fibers are isotropy grous, they are properly embedded.*

### Lecture 18: Vector fields

Wed 15 Oct 2025 12:44

Recall that given a manifold $M$, a smooth vector field is a section $V : M \to TM$ of the projection of the tangent bundle $\pi : TM \to M$. The field $V$ at a point $p \in M$ then takes the form

$$V_p = V_p^i \partial_i|_p.$$

We can then define vector field in ways like

$$x \mapsto x^i \partial_i|_x.$$

If we choose a different chart on $M$, we could obtain a different coordinate function $\partial_i$ and get new fields.

### Lecture 19: idk

Fri 17 Oct 2025 12:20

Let $F : M \to N$ be a smooth map. For each $p \in M$, we get an induced map of tangent spaces $\mathrm{d}F_p : T_pM \to T_{F(p)}N$. However, if $V$ is a smooth vector field on $M$, we cannot in general define $\mathrm{d}FV$ as a smooth vector field on $N$.

**Definition 3.0.22.** Let $F : M \to N$ be smooth, and let $X : M \to TM$ and $Y : N \to TN$ be vector fields. Then $X$ and $Y$ are $F$-related if and only if for all $p \in M$, $\mathrm{d}F_p(V_p) = Y_p$, where of course $X_p = X|T_pM$, and the same for $Y$.

This tells us that $X(f \circ F) = Y(f) \circ F$.

**Proposition 3.0.23.** *Let $F : M \to N$ be a diffeomorphism. Then for all vector fields $X : M \to TM$, there exists a unique vector field $Y : N \to TN$ which is $F$-related to $X$.*

**Definition 3.0.24.** Let $X, Y$ be smooth vector fields on $M$. Define the Lie bracket $[X, Y] \coloneqq (X \circ Y) - (Y \circ X)$.

**Definition 3.0.25.** A Lie algebra is a ($\mathbb{R}$-)vector space $\mathfrak{g}$ together with a Lie bracket $[-,-] : \mathfrak{g} \times \mathfrak{g} \to \mathfrak{g}$ which is antisymmetric, bilinear, and satisfies the Jacobi identity. A morphism of Lie algebras is a linear map $f : \mathfrak{g} \to \mathfrak{h}$ such that $f[X, Y]_{\mathfrak{g}} = [f(X), f(Y)]_{\mathfrak{h}}$.

To any manifold $M$ is assocaited a Lie algebra $\mathfrak{X}$ of smooth vector fiekds, with Lie bracket given by definition 3.0.24. Moreover, the Lie bracket is natural in the following sense: suppose $F : M \to N$ is smooth, and suppose $X_1, X_2 \in \mathfrak{X}(M)$, $Y_1, Y_2 \in \mathfrak{X}(N)$ are smooth vector fields with $X_i$ $F$-related to $Y_i$. Then the Lie bracket $[X_1, X_2]$ is also $F$-related to $[Y_1, Y_2]$. In particular, diffeomorphisms $F : M \cong N$ induce Lie algebra homeomorphisms. A corollary of this is if $S \subseteq M$ is immersed, then the Lie bracket of two vector fields tangent to $S$ is also tangent to $S$.

**Proposition 3.0.26.** *Given a Lie group $G$, there exists a natural finite dimensional Lie algebra $\mathfrak{g}$ such that $\mathfrak{g} \subseteq \mathfrak{X}(G)$ and $[\mathfrak{g}, \mathfrak{g}] \subseteq \mathfrak{g}$.*

**Definition 3.0.27.** Given a Lie group $G$ and a vector field $V \in \mathfrak{X}(G)$, we say $V$ is a *left-invariant* vector field on $G$ if and only if $L_{g*}V = V$ for all $g \in G$. Right invariance is defined similarly. The Lie algebra of proposition 3.0.26 is given by the set of left-invariant vector fields.

## Lecture 20: The Lie algebra of a Lie group

Mon 20 Oct 2025 12:21

## Lecture 21: Cotangent bundle

Mon 03 Nov 2025 12:28

Let $T_p^*M$ be the dual space $\mathrm{Hom}(T_pM, \mathbb{R})$. This has a basis $\mathrm{d}x^i|_p$ which acts by

$$\mathrm{d}x^i|_p(\partial_j|_p) = \delta_i^j.$$

We call this the cotangent space at $p$, and elements are tangent covectors. The cotangent bundle

$$T^*M \coloneqq \coprod_{p\in M} T_p^*M$$

has a smooth structure analogous to $TM$, and has a smooth projection $\pi : T^*M \to M$ via $(p, \omega) \mapsto p$. Smooth sections are called covector fields, and we write $\mathfrak{X}^*(M)$ for the set of covector fields. There is a linear map, called the *gradient*

$$\mathrm{d} : C^\infty(M) \to \mathfrak{X}^*(M)$$

which maps functions $f$ to vector fields $\mathrm{d}f$ which act on tangent vectors $V_p \in T_pM$ via

$$(\mathrm{d}f)_p(V_p) = V_p(f).$$

In coordinates,

$$\mathrm{d}f = \frac{\partial f}{\partial x^\mu}\mathrm{d}x^\mu.$$

Given a smooth map $F : M \to N$, the pushforward $F_{*p} : T_pM \to T_{F(p)}N$ is a covariant action. Since dualizing is contravariant, this induces a *pullback*

$$F_p^* : T_{F(p)}^*N \to T_p^*M.$$

This is defined as follows: given a covector $\alpha \in T_{F(p)}^*N$ and a vector $V_p \in T_pM$,

$$(F_p^*\alpha)(V_p) = \alpha(F_{*p}V_p).$$

Indeed, $F_{*p}V_p \in T_{F(p)}N$, so $\alpha$ (which we write $\alpha_{F(p)}$ usually) can act on it. This is nice because we can pull back covector fields. Given $\omega \in \mathfrak{X}^*(N)$ and a smooth $F : M \to N$, then define $F^*\omega$ by

$$(F^*\omega)_p(V_p) = (F_p^*\omega_{F(p)})(V_p) = \omega_{F(p)}(F_{*p}V_p).$$

In other words, $F^* : \mathfrak{X}^*(N) \to \mathfrak{X}^*(M)$.

**Lemma 3.0.28.** *Let $G : M \to N$ be smooth and $f : N \to \mathbb{R}$ smooth. Then*

$$G^*(\mathrm{d}f) = \mathrm{d}(f \circ G).$$

*If $\omega \in \mathfrak{X}^*(N)$, then*

$$G^*(f\omega) = (f \circ G)(G^*\omega).$$

*Proof.* By definition,

$$G^*(\mathrm{d}f)_p(X_p) = (\mathrm{d}f)_{G(p)}(G_{*p}X_p) = (G_{*p}X_p)(f) = X_p(f \circ G) = \mathrm{d}(f \circ G)_p(X_p).$$

I don't even know what the second implication is stating. ■

**Remark 3.0.29.** There is a functor $T^* : \mathcal{M}\mathrm{an}_*^{\mathrm{op}} \to \mathbb{R}\text{-}\mathcal{V}\mathrm{ect}$, where $\mathcal{M}\mathrm{an}_*$ is the category of based manifolds. This is given by $(M, p) \mapsto T_p^*M$, and $F \mapsto F_p^*$.

## 3.1 Line integrals of covector fields

Let $J = [a, b] \subseteq \mathbb{R}$, and let $\omega \in \mathfrak{X}^*(J)$. This can be given by $\omega = f(t)\mathrm{d}t$, where $\mathrm{d}t$ is the single basis element in our chart and $f \in C^\infty(J)$. We can define

$$\int_J \omega = \int_a^b f(t)\,\mathrm{d}t.$$

**Proposition 3.1.1.** *Let $\omega$ be a smooth covector field on an interval $J$, and let $\varphi : [c, d] \to [a, b]$ be an increasing diffeomorphism. Then*

$$\int_{[c,d]} \varphi^*\omega = \int_{[a,b]} \omega.$$

*Proof.*

$$\begin{aligned}(\varphi^*\omega)_s &= \varphi^*(f(t)\mathrm{d}t)\\ &= (\varphi^* f)\mathrm{d}\varphi^* t\\ &= f(\varphi(s))\mathrm{d}\varphi(s)\\ &= f(\varphi(s))\varphi'(s)\mathrm{d}s.\end{aligned}$$

Here $\varphi(s) = t$. We then recover

$$\int_{[c,d]} f(\varphi(s))\varphi'(s)\,\mathrm{d}s = \int_{[a,b]} f(t)\,\mathrm{d}t$$

from the chain rule. ■

**Definition 3.1.2.** Let $M$ be a smooth manifold, and $\gamma : J \to M$ a continuous and piecewise smooth curve in $M$ with $J$ an interval. Let $\omega \in \mathfrak{X}^*(M)$. Define the line integral of $\omega$ over $J$ as

$$\int_\gamma \omega := \int_J \gamma^*\omega.$$

This definition makes sense since $\gamma^*\omega$ is a smooth covector field on $J$.

Lee 11.33 says any two points on a smooth manifold can be connected with a piecewise smooth curve.

**Proposition 3.1.3.** *The integral over a curve $\gamma$ is $\mathbb{R}$-linear. If $\gamma$ is constant than the integral over it is 0. If $\gamma : [a, b] \to M$ can be split into piecewise smooth curves $\gamma_1 : [a, c] \to M$ and $\gamma_2 : [c, b] \to M$, then*

$$\int_\gamma \omega = \int_{\gamma_1} \omega + \int_{\gamma_2} \omega.$$

*If $F : M \to N$ is smooth, then*

$$\int F^*\omega = \int_{F\circ\gamma} \omega.$$

**Remark 3.1.4** (Author's remarks)**.** The tangent space at $p$ is functorial $(p, M) \mapsto T_pM$ and $F \mapsto F_{*p}$. Then $\mathrm{Hom}(-, \mathbb{R})$ is a functor mapping $T_pM$ to $T_p^*M$, so the definition of $F_p^*$ can be given more transparently. Given $\omega \in T_p^*M$, define $F_p^*\omega \coloneqq \omega \circ F_{*p}$. This is equivalent to what was given, since for any vector $V \in T_pM$,

$$(F_p^*\omega)(V) = (\omega \circ F_{*p})(V) = \omega(F_{*p}V).$$

The gradient $\mathrm{d} : C^\infty(M) \to \mathfrak{X}^*(M)$ can be interpreted as the map $f \mapsto \mathrm{Ev}_f$, since for any $p \in M$ and any tangent vector $V \in T_pM$,

$$(\mathrm{d}f)_p(V) = V(f) = \mathrm{Ev}_f(V).$$

In other words, $\mathrm{d}f = \mathrm{Ev}_f$. And of course the gradient has the coordinate representation

$$\mathrm{d}f = \frac{\partial f}{\partial x^\mu}\mathrm{d}x^\mu \implies \mathrm{d}x^\mu = \frac{\partial x^\mu}{\partial (x')^\nu}\mathrm{d}(x')^\nu.$$

This agrees with the transformation law given in GR, from which we obtain the transformation law for covectors

$$\omega'^\mu = \omega_\nu \frac{\partial x^\nu}{\partial (x')^\mu}.$$

Furthermore, the pullback for vector fields is just given by coproducting along. Given $\omega \in \mathfrak{X}^*(N)$ and $F : M \to N$ smooth, and given $p \in M$ and $V \in T_pM$, define $F^* : T^*_{F(p)}N \to T_p^*M$ by $F^*\omega = \omega \circ F_*$ pointwise in the sense that

$$(F^*\omega)_p = F_p^*\omega_{F(p)} = \omega_{F(p)} \circ F_{*p}.$$

Acting on a tangent vector, this is the same as what was given:

$$(F^*\omega)_p(V) = (\omega_{F(p)} \circ F_{*p})(V) = \omega_{F(p)}(F_{*p}V).$$

Now let $J \subseteq \mathbb{R}$ be an interval. We said earlier that any covector field $\omega$ on $J$ can be written as $\omega = f(t)\mathrm{d}t$ for $f \in C^\infty(J)$. This an abuse of notation; we should write that for any given $t \in J$, the component of $\omega$ at $t$ is given as $\omega_t = f(t)\mathrm{d}t$.

## Lecture 22: Covector fields

Wed 05 Nov 2025 12:20

Recall: given a path $\gamma : [a, b] \to M$ in a manifolds and a covector field $\omega \in \mathfrak{X}^*(M)$, we can define the *line integral* of $\omega$ over $\gamma$ by

$$\int_\gamma \omega = \int_a^b \gamma^*\omega.$$

In vector calculus, the line integral of a vector field $V : \mathbb{R}^m \to \mathbb{R}^n$ over a path $\gamma : [a, b] \to \mathbb{R}^n$ was given by

$$\int_\gamma V \cdot \mathrm{d}\mathbf{s} = \int_a^b V(\gamma(t))\frac{\partial \gamma}{\partial t}\,\mathrm{d}t.$$

It would be interesting if our formula basically became the same. The reason vector calc used vectors instead of covectors was because we had a nice inner product. On manifolds, we don't, so we have to use covectors and do more work.

Given a vector space $V$, we can define a bilinear map $g : V \times V \to \mathbb{R}$ by $g(v_1, v_2) = v_1 \cdot v_2$ the dot product. We can then define a map $g^\flat : V \to V^*$ by $v \mapsto g(v, -)$. This is linear and has trivial kernel, so its an isomorphism of vector spaces.

Let's move on to considering conservative fields. In calculus 3, a vector field $V$ was called conservative if it had a potential function $V = \nabla f$ for some $f$, and it was path independent if and only if it was conservative.

**Proposition 3.1.5.** *If $\gamma : [a, b] \to M$ is piecewise smooth and $\omega$ is a covector field on $M$, then*

$$\int_\gamma \omega = \int_a^b \omega_{\gamma(t)}(\gamma'(t))\,\mathrm{d}t.$$

*This reduces this to just a calculus 1 integral.*

**Proposition 3.1.6.** *Let $M$ be a smooth manifold, $\omega \in \mathfrak{X}^*(M)$, and $\gamma : [a, b] \to M$ piecewise smooth. Let $\widetilde{\gamma}$ be a* forwards representation *of $\gamma$. Then*

$$\int_{\widetilde{\gamma}} \omega = \int_\gamma \omega.$$

*If $\widetilde{\gamma}$ is a* backwardsd reparameterization*, then*

$$\int_{\widetilde{\gamma}} \omega = -\int_\gamma \omega.$$

**Definition 3.1.7.** Given $\gamma : [a, b] \to M$ a piecewise smooth curve, a forwards reparameterization is a reparameterization which is monotonic, so if $\gamma$ is in terms of $t$ and we change $t \to s$, then as $t$ increases, $s$ increases, and vice versa. A backwards reparameterization is the opposite.

**Theorem 3.1.8.** *Let $M$ be a smooth manifold, $\gamma : [a, b] \to M$ piecewise smooth, and $f : M \to \mathbb{R}$ a smooth function. Then recalling that $\mathrm{d}f \in \mathfrak{X}^*(M)$,*

$$\int_\gamma \mathrm{d}f = (f \circ \gamma)(b) - (f \circ \gamma)(a).$$

*Proof.* By our definition of d and proposition 3.1.5,

$$\int_\gamma \mathrm{d}f = \int_a^b \mathrm{d}f_{\gamma(t)}(\gamma'(t))\,\mathrm{d}t = \int_a^b (f \circ \gamma)'(t)\,\mathrm{d}t.$$

Applying the fundamentl theorem of calculus recovers the statement. ∎

**Corollary 3.1.9.** *If $\gamma(a) = \gamma(b)$, then*

$$\int_\gamma \mathrm{d}f = 0.$$

**Definition 3.1.10.** Let $\omega \in \mathfrak{X}^*(M)$. Then $\omega$ is *exact* if and only if $\omega = \mathrm{d}f$ for some $f \in C^\infty(M)$. We say $f$ is the *potential function* of $\omega$. We say $\omega$ is *conservative* if and only if

$$\int_\gamma \omega = 0$$

for all piecewise smooth *closed* curves (loops) $\gamma$.

In calculus, exactness and conservativeness were identical. On manifolds, we have aleady shown that exact $\implies$ conservative. Is the converse true?

**Proposition 3.1.11.** *A covector field $\omega \in \mathfrak{X}^*(M)$ is conservative if and only if*

$$\int_\gamma \omega = \int_{\widetilde{\gamma}} \omega$$

*for all $\gamma, \widetilde{\gamma}$ starting and ending at the same points.*

*Proof.* Using proposition 3.1.3 and proposition 3.1.6, form a closed loop be reparameterizing one of them backwards and adding them together. ■

**Theorem 3.1.12.** *A smooth covector field $\omega$ on $M$ is conservative if and only if it is exact.*

*Proof.* We already have one direction, so let $\omega$ be conservative. We can assume $M$ is connected by taking a connected component. Let $\gamma : [a, b] \to M$ be piecewise smooth, and write

$$\int_{\gamma(a)}^{\gamma(b)} \omega = \int_\gamma \omega.$$

Since $\omega$ is conservative, this notation is well-defined. Pick a basepoint $p_0 \in M$, and define

$$f(q) \coloneqq \int_{p_0}^{q} \omega.$$

We claim that $\mathrm{d}f = \omega$. Choose some $q_0 \in M$, and choose a chart $(U, \varphi)$ of $q_0$. Write $x^i$ for the coordinates. The blackboard got messy and I got lost but it follows. Basically you write it in coordinates then use the same argument we did in calculus 3 to show the proof. I don't remember the calculus three argument... ■

## Lecture 23: Tensors

Mon 10 Nov 2025 12:23

We are trying to define a bilinear form $g_p : T_pM \times T_pM \to \mathbb{R}$ which is smooth and positive-definite, like a smooth inner product. It will be useful to start by defining a smooth map $T_pM \otimes T_pM \to \mathbb{R}$ and then precomposing with tensor.

**Notation 3.1.13.** Define

$$T^k T_p M \coloneqq (T_p M)^{\otimes k},$$
$$T^k T_p^* M \coloneqq \left(T_p^* M\right)^{\otimes k},$$
$$T^{(k,\ell)} T_p M \coloneqq T^k T_p M \otimes T^\ell T_p^* M.$$

Then define

$$T^{(k,\ell)} TM \coloneqq \coprod_{p \in M} T^{(k,\ell)} T_p M.$$

We also use the notation $T^{(0,k)}TM = T^k T^* M$ and $T^{(k,0)}TM = T^k TM$. Finally, we denote the tensor algebra on a $\mathbb{R}$-linear space $V$ by $\mathbb{T}V$. The professor uses $\mathcal{T}^{(k,\ell)}TM$ instead.

The tensor product of duals $V^* \otimes V^*$, then this is naturally isomorphic to the set $L(V, V; \mathbb{R})$ of bilinear forms $V \times V \to \mathbb{R}$.

**Definition 3.1.14.** A *covariant* $k$-tensor in a $\mathbb{K}$-vector space $V$ is an element of $(V^*)^{\otimes k}$. This terminology is ancient and terrible and some books flip this with contravariant so we will prefer to just call this a $k$-cotensor or a $k$-multilinear form on $V$. Similarly, a *contravariant* $k$-tensor is an element of $V^{\otimes k}$. We will just call this a $k$-tensor.

**Definition 3.1.15.** A *Riemannian metric* on a manifold $M$ is a smooth section of $T^2 T^* M$ which is symmetric and positive definite.

## Lecture 24: Riemannian metrics

Fri 14 Nov 2025 12:19

**Definition 3.1.16.** A Riemannian metric $g$ on a smooth manifold $M$ is a smooth section of $T^*M \otimes T^*M$ such that each $g_p : T_pM \otimes T_pM \to \mathbb{R}$ is symmetric and positive definite, meaning $g(v,v) = 0$ iff $v = 0$.

In coordinates we write

$$g_p = \sum_i \sum_j g_{ij} \mathrm{d}x^i \otimes \mathrm{d}x^j.$$

An example is $g_{ij} = \delta_i^j$, which makes

$$g_p(W, V) = V^i W^i = V \cdot W.$$

**Proposition 3.1.17.** *Every smooth manifold admits a Riemannian metric.*

We write $\overline{g}$ for the standard Euclidean metric.

**Proposition 3.1.18.** *Cover $M$ by charts $\{(U_\alpha, \varphi_\alpha)\}_{\alpha \in A}$. Then $g_\alpha = \varphi_\alpha^* \overline{g}_\alpha$.*

We can define the length $L(\gamma)$ of a curve $\gamma$ as

$$\int_a^b \|\dot{\gamma}(t)\| \, \mathrm{d}t.$$

Of course $\|\gamma(t)\| = \sqrt{g_{\gamma(t)}(\dot{\gamma}, \dot{\gamma})}$.

**Definition 3.1.19.** Let $(M, g)$ and $(N, h)$ be Riemannian manifolds. A diffeomorphism $F : M \to N$ is called an *isometry* if $F^*h = g$. If every $p \in M$ has an open neighborhood isometric to an open set of $N$, then $(M, g)$ is *locally* isometric to $(N, h)$.

**Definition 3.1.20.** A Riemannian $n$-manifold is *flat* if and only if it is locally isometric to $\mathbb{R}^n$.

## Lecture 25: More metrics

Mon 17 Nov 2025 12:32

Let $i : S^2 \to \mathbb{R}^3$ be the inclusion. We define the metric $g^\circ \coloneqq i^*\overline{g}$ on the sphere. If we work it out then it looks like

$$g^\circ = \left(1 - v^2\right) \mathrm{d}u^2 + \left(1 - u^2\right) \mathrm{d}v^2 + 2uv \mathrm{d}u \mathrm{d}v.$$

If we don't write the tensor product between covectors, then that is understood to be the wedge product

$$\mathrm{d}u\mathrm{d}v \coloneqq \mathrm{d}u \wedge \mathrm{d}v = \frac{\mathrm{d}u \otimes \mathrm{d}v + \mathrm{d}v \otimes \mathrm{d}u}{2}.$$

Given a Riemannian manifold $(M, g)$, there is an isomorphism $g_p^\flat : T_pM \to T_p^*M$ given by the linear map $g_p^\flat(v)(v) = 1$. The kernel is trivial by definition, so it is indeed an isomorphism. Turns out this is even a diffeomorphism, and if it commutes with the projections then it is an isomorphism of vector bundles.

## Lecture 26: Forms

Fri 21 Nov 2025 12:43

The exterior algebras of $k$-tensors $\Lambda^k T_p^* M$ assemble into a vector bundle $\Lambda^k T^* M$. Sections of this bundle are called $k$-differential forms or $k$-forms, and the set of sections are denoted

$$\Omega^k(M) \coloneqq \Gamma(\Lambda^k T^* M).$$

We can then define the exterior algebra

$$\Omega^\bullet(M) \coloneqq \bigoplus_{i=0}^{\infty} \Omega^i$$

with wedge product given pointwise by, for any $\omega \in \Omega^k(M)$ and $\psi \in \Omega^\ell(M)$, by antisymmetrizing the tensor product $\omega \otimes \psi$. We can as usual pull back $k$-forms $\omega$ by smooth maps $F : M \to N$:

$$(F^*\omega)_p(v_1, \ldots, v_k) = \omega_{F(p)}(F_{*p}v_1, \ldots, F_{*p}v_k).$$

This is linear, preserves the wedge product, and in coordinates

$$F^*\left(\sum_I \omega_I \mathrm{d}x^I\right) = \sum_i (\omega_i \circ F)\mathrm{d}(x^{i_1} \circ F) \wedge \cdots \wedge \mathrm{d}(x^{i_k} \circ F).$$

Locally, $k$-forms look like $\omega = \sum_i \omega_i \mathrm{d}x^{i_1} \wedge \cdots \wedge \mathrm{d}x^{i_k}$. We typically write $\omega = \sum_I \omega_I \mathrm{d}x^I$ for simplicity.

**Definition 3.1.21.** The exterior derivative $\mathrm{d} : \Omega^k(M) \to \Omega^k(M)$ is the map given by

$$\omega = \sum_I \omega_I \mathrm{d}x^I \mapsto \sum_I \left(\sum_{j=1}^n \frac{\partial \omega_I}{\partial x^j}\mathrm{d}x^j\right) \wedge \mathrm{d}x^I.$$

We write the coefficient as $\mathrm{d}\omega_I$.

If $k = 0$ then $\Omega^0(M) = C^\infty(M)$, and we obtain $\mathrm{d}f = \partial_{x^i} f \mathrm{d}x^i$ which is the gradient. For $k = 1$, we have $\Omega^1(M) = \mathfrak{X}^*(M)$, yielding

$$\mathrm{d}\omega = \sum_{j=1}^n (\partial_{x^j}\omega_i - \partial_{x^i}\omega_j)\,\mathrm{d}x^i \wedge \mathrm{d}x^j.$$

This definition requires charts which kinda sucks. Grinding out the chain rule and product rule allows us to find

**Proposition 3.1.22.** *1.* $\mathrm{d}$ *is* $\mathbb{R}$*-linear.*

*2. If $\omega$ is a $k$-form and $\eta$ is a 1-form on an open subset $U \subseteq \mathbb{R}^n$ or $\mathbb{H}^n$, then*

$$\mathrm{d}(\omega \wedge \eta) = \mathrm{d}\omega \wedge \eta + (-1)^k \omega \wedge \mathrm{d}\eta.$$

*3.* $\mathrm{d}^2 = 0$.

*4.* $\mathrm{d}$ *commutes with pullbacks: $F^*(\mathrm{d}\omega) = \mathrm{d}F^*(\omega)$ for any smooth $F : M \to N$ and $k$-form $\omega$.*

We can actually generalize d to a map $\Omega^k \to \Omega^{k+1}$ in such a way that the first 3 properties still hold, and on 0-forms $f : M \to \mathbb{R}$ it acts by $\mathrm{d}f(X) = X(f)$.

**Definition 3.1.23.** We say a $k$-form is *closed* iff $\mathrm{d}\omega = 0$, and exact iff $\omega = \mathrm{d}\eta$ for some $\eta$. Note that exact $\implies$ closed by $\mathrm{d}^2 = 0$.

**Example 3.1.24.** If we try to compute d on $\mathbb{R}^3$ and those manifolds we get $\nabla$, $\nabla\times$, and $\nabla\cdot$.

## Lecture 27: More forms

Mon 24 Nov 2025 12:19

Recall that a differential form is a section of $\Lambda^k(T^*M)$. We denote the set of $k$-forms by $\Omega^k(M)$. There is a map $\mathrm{d} : \Omega^k(M) \to \Omega^{k+1}(M)$ called the *exterior derivative* which we defined in coordinates. We then showed the fundamental properties of linearity, $\mathrm{d}^2 = 0$, d commutes with pullbacks, and for all 1-forms $\eta$ and $k$-forms $\omega$,

$$\mathrm{d}(\omega \wedge \eta) = \mathrm{d}\omega \wedge \eta + (-1)^k \omega \wedge \mathrm{d}\eta.$$

Define

$$\Omega^k(M) \coloneqq \coprod_{p \in M} \Lambda^k(T_p^* M).$$

Passing to

$$\Omega^\bullet(M) \coloneqq \bigoplus_{k \geq 0} \Omega^k(M)$$

yields an exterior algebra with wedge product given by $(\omega \wedge \eta)_p = \omega_p \wedge \eta_p$.

We claim that d subsumes the curl and gradient and divergence in the following sense. Define $\beta(V) \coloneqq (\mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z)(V, -, -)$. Then the diagram commutes:

$$\begin{tikzcd}
C^\infty(\mathbb{R}^3) \arrow[r, "\nabla"] \arrow[d, equal] & \mathfrak{X}(\mathbb{R}^3) \arrow[r, "\mathrm{curl}"] \arrow[d, "\overline{g}^\flat"] & \mathfrak{X}(\mathbb{R}^3) \arrow[r, "\mathrm{div}"] \arrow[d, "\beta"] & C^\infty(\mathbb{R}^3) \arrow[d, "*"] \\
C^\infty(\mathbb{R}^3) \arrow[r, "\mathrm{d}"'] & \Omega^1(\mathbb{R}^3) \arrow[r, "\mathrm{d}"'] & \Omega^2(\mathbb{R}^3) \arrow[r, "\mathrm{d}"'] & \Omega^3(\mathbb{R}^3)
\end{tikzcd}$$

**Proposition 3.1.25.** *Let $M$ be a smooth $n$-manifold, $\omega$ a $k$-form, and $V_1, \ldots, V_{k+1}$ vector fields. Then*

$$(\mathrm{d}\omega)(V_1, \ldots, V_{k+1}) = \sum_{i=1}^{k+1} (-1)^i V_i(\omega(V_1, \ldots, \hat{V}_i, \ldots, V_{k+1})) - \sum_{1 \leq i < j \leq n} (-1)^{i+j} \omega([V_i, V_j], V_1, \ldots, \hat{V}_i, \hat{V}_j, \ldots, V_{k+1}).$$

*The hat of course denotes that we omit the argument.*

We can define the Lie derivative on forms in the exact same way as for tensor fields. Given a vector field $V$, a $k$-form $\eta$ and an $\ell$-form $\psi$,

$$\mathcal{L}_V \omega = \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} (\theta_t^* \omega)$$

$$\mathcal{L}_V(\omega \wedge \psi) = (\mathcal{L}_V \omega) \wedge \psi + \omega + (\mathcal{L}_V \psi).$$

**Definition 3.1.26.** Let $V$ be a finite vector space. Define for any $X \in V$ the operator $\iota_X : \Lambda^k V^* \to \Lambda^{k-1} V^*$ by $\omega \mapsto \omega(X, -, \ldots, -)$. We call this the *interior product* with respect to $X$.

**Proposition 3.1.27.** *Ler $X, Y \in V$.*

1. $\iota_X \circ \iota_Y = -\iota_{Y \circ X}$.
2. $\iota_X(\omega \wedge \psi) = (\iota_X \omega) \wedge \psi + (-1)^k \omega \wedge \iota_X \psi$.
3. $\iota_X$ *is an odd derivation.*

**Proposition 3.1.28.** *Cartan's formula:*

$$\mathcal{L}_X = \mathrm{d} \circ \iota_X + \iota_X \circ d : \Omega^k(M) \to \Omega^k(M).$$

*Proof.* The proof goes by showing they agree on functions and 1-forms. Then by the uniqueness proofs we can do with the product rule on Lie derivatives, it follows. ■

**Construction 3.1.29.** Since $\mathrm{d}^2 = 0$ and since the set of *sections* is a vector space, there is a cochain complex

$$\Omega^0(M) \xrightarrow{\mathrm{d}_p} \Omega^1(M) \xrightarrow{\mathrm{d}_p} \Omega^2(M) \xrightarrow{\mathrm{d}_p} \cdots$$

The kernel is the set of closed forms and the image is the set of exact forms. Thus the cohomology, called the De Rahm cohomology, can be given as

$$H^k_{\mathrm{dh}}(M) \coloneqq \frac{Z^k(M)}{B^k(M)},$$

for $Z^k(M)$ the set of closed $k$-forms and $B^k(M)$ the set of exact $k$-forms. Note that $H^\bullet_{\mathrm{dh}} = 0$ if and only if closed and exact $k$-forms coencide on $M$.

## Lecture 28: Integrals of forms

Fri 05 Dec 2025 12:21

Give $\mathbb{R}^n$ the standard orientation, and let $\omega \in \Omega^n_c(\widetilde{U})$, where $\Omega_c$ is compactly supported forms. Then $\omega = f(x^1, \ldots, x^n)\mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^n$. To integrate this in $\mathbb{R}^n$, we just do

$$\int_{\widetilde{U}} \omega = \int_{\widetilde{U}} f(x^1, \ldots, x^n)\, \mathrm{d}x^1 \cdots x^n.$$

We would like to generalize this. Let $(M, \mathrm{or})$ be an oriented $n$-manifold and $\omega \in \Omega^n(M)$ which is compactly supported in some chart $(U, \varphi)$ which is either positively or negatively oriented. Define

$$\int_M \omega \coloneqq \pm \int_U \varphi^* \omega,$$

with the positive sign for the positive orientation and the negative sign for the negative orientation. It is a simple theorem that this is well-defined in the sense that it is invariant under choice of chart.

To finally define integrals over the entire manifold, let $M$ be an $n$-manifold with orienation, and $\omega$ a compactly supported $n$-form on $M$. Let $\{U_i\}$ be a finite open cover of $\operatorname{supp} \omega$ by charts either all positively or negatively oriented, and let $\{\psi_i\}$ be a subordinate smooth partition of unity. Define

$$\int_M \omega \coloneqq \sum_i \int_M \psi_i \omega.$$

Here, juxtaposition denotes *pointwise multiplication.*

**Proposition 3.1.30.** *This definition is well-defined. Moreover, it is $\mathbb{R}$-linear, is orientation reversing (if $-M$ is $M$ with opposite orientation, then:*

$$\int_{-M} \omega = -\int_M \omega,$$

*if $\omega$ is a positively oriented form, then $\int_M \omega > 0$, and if $F : M \to N$ is an orientation preserving or reversing diffeomorphism, then*

$$\int_M \omega = \pm \int_N F^* \omega,$$

*with $-$ if $F$ is orientation-reversing.*

**Theorem 3.1.31** (Stokes)**.** *Let $M$ be a smooth oriented $n$-manifold with boundary, and let $\omega \in \Omega_c^{n-1}(M)$. Then*

$$\int_M \mathrm{d}\omega = \int_{\partial M} \omega.$$

**Remark 3.1.32.** We implicitly interpret the $\omega$ on the right as $\iota_{\partial M}^* M$, and induce the usual orientation on the boundary. If $M$ doesn't have boundary, then $\partial M = \varnothing$, so the integral on the right vanishes.

# Chapter 4

# Orientations

**Lecture 29: Personal Review**

Mon 12 Jan 2026 13:59

This section is me reviewing in preparation for courses next semester. I intend to review starting from orientations since they were very confusing, and will continue through integration and Stokes, de Rham cohomology, the de Rham theorem, and probably distributions, foliations, and the exponential. I think I will omit quotient manifolds and symplectic manifolds for now.

## 4.1 Orientations of vector spaces

The spaces $\mathbb{R}^n$ has a canonical basis $\{e_i = \delta_{ij}\}_{1 \le i \le n}$. This induces the conventionally standard "right-handed" orientation on $\mathbb{R}^n$. The notion of "right-handedness" can be formalized as follows: the transition matrix $e_i^j$ of the basis has positive determinant. This leads us to the following generalization to vector spaces with no canonical basis.

**Definition 4.1.1.** Let a nontrivial finite $\mathbb{R}$-vector space $V$ have bases $\{e_i\}, \{\widetilde{e}_i\}$. We say these bases are *consistently oriented* if the transition matrix $B_i^j$ given by

$$e_i = B_i^j \widetilde{e}_j$$

has positive determinant.

**Proposition 4.1.2.** *Being consistently oriented is an equivalence relation.*

*Proof.* The transition matrix $e_i \to e_i$ is 1 which has determinant 1. Let $B$ be a transition matrix $e_i \to \widetilde{e}_i$ of positive determinant; since it has nonzero determinant it has an inverse which clearly satisfies the properties of a transition matrix $\widetilde{e}_i \to e_i$. Since det preserves multiplication,

$$\det B^{-1} = \frac{1}{\det B} > 0$$

because $\det B > 0$. Finally, let $B : e_i \to \widetilde{e}_i$ and $C : \widetilde{e}_i \to \hat{e}_i$ be transition matrices of positive determinant. Then the transition matrix $e_i \to \hat{e}_i$ is simply $CB$ which has determinant $\det CB = \det C \det B > 0$ since both $\det C, \det B > 0$. ∎

**Definition 4.1.3.** Let $V$ be a nontrivial finite $\mathbb{R}$-vector space. Define an *orientation* of $V$ as an equivalence class of bases under the relation given by proposition 4.1.2.

**Notation 4.1.4.** Given a basis $(e_1, \dots, e_n)$ for $V$, we write the orientation it determines as $[e_1, \dots, e_n]$, or simply $[e_i]$. We denote the opposite orientation by $-[e_i]$, which is given by $-[e_i] = [-e_i]$.

An *oriented* vector space is a vector space together with a choice of orientation. Any basis in the orientation is said to be *positively oriented*, and otherwise it is said to be *negatively oriented*. For the special case $V = 0$, we define an orientation to be a choice of a number $\pm 1$.

For example, $\mathbb{R}^n$ comes with the standard orientation $[\delta_i^j]$.

**Proposition 4.1.5.** *Let $V$ be a $\mathbb{R}$-vector space with dimension $n$. Each nonzero $\omega \in \Lambda^n(V^*)$ determines an orientation $\mathcal{O}_\omega$ of $V$ as follows. If $n \geq 1$, then $\mathcal{O}_\omega$ is the set of bases $\{e_i\}_{1 \leq i \leq n}$ such that $\omega(e_1, \dots, e_n) > 0$; if $n = 0$, then $\mathcal{O}_\omega = 1$ if $\omega > 0$, and $\mathcal{O}_\omega = -1$ if $\omega < 0$. Two nonzero alternating $n$-covectors determine the same orientation if and only if one is a positive multiple of the other.*

*Proof.* We define $\Lambda^0(0^*)$ to be a real number, so this follows immediately. Take $n \geq 1$. Let $\omega \in \Lambda^n(V^*)$ be nonzero, and define $\mathcal{O}_\omega$ as in the proposition. We must show it is indeed an orientation. It suffices to show it is precisely an equivalence class.

Let $\{e_i\}, \{\widetilde{e}_j\}$ be bases for $V$ and $B$ the transition matrix $e_i \to \widetilde{e}_i$:

$$\widetilde{e}_j = B_j^i e^i.$$

Recall that for any linear map $T : V \to V$ and any vectors $\{v_i\}$ and any alternating $n$-covector $\eta$,

$$\eta(Tv_1, \dots, Tv_n) = (\det T)\eta(v_1, \dots, v_n).$$

Applying this yields

$$\omega(\widetilde{e}_1, \dots, \widetilde{e}_n) = \omega(Be_1, \dots, Be_n) = (\det B)\omega(e_1, \dots, e_n).$$

The bases are consistently oriented if and only if $\det B > 0$, and we see that this happens if and only if $\omega(\widetilde{e}_1, \dots, \widetilde{e}_n)$ and $\omega(e_1, \dots, e_n)$ have the same sign. The last statement follows immediately by noting that the sign is determined by the determinant of the transition matrices. ∎

If $V$ is oriented and $\omega \in \Lambda^n(V^*)$ determines the orientation of $V$, we say $\omega$ is positively oriented. For example, $\bigwedge_i e_i$ is positively oriented on $\mathbb{R}^n$.

Recall that space $\Lambda^n(V^*)$ is 1-dimensional[1]. The second statement of proposition 4.1.5 shows that choosing a nonzero component is the same as choosing a connected component of $\Lambda^n(V^*) \setminus \{0\}$. This also works for $V = 0$, explaining why we defined the orientation of 0 as we did.

## 4.2 Orientations of manifolds

The notion of an oriented vector space generalizes to manifolds quite immediately.

**Definition 4.2.1.** Let $M$ be a manifold. A *pointwise orientation* is a choice of an orientation for each tangent space. If $\{E_i : (U \subseteq M) \to TM\}$ is a local frame (not necessarily smooth!) for $M$, we say $\{E_i\}$ is *positively oriented* if $\{E_1|_p, \dots, E_n|_p\}$ is a positively oriented basis for each $T_pM$; negatively oriented frames are defined analogously. A pointwise orientation is *continuous* if each point in $M$ is in the domain of some local frame. An *orientation* for $M$ is then a continuous pointwise

[1]The basis for $\Lambda^k(V)$ is given by choosing an *ordered* basis and taking $k$-fold wedge products with strictly increasing indicies. Take $n = 1$; there is only one wedge product with increasing indicies.

orientation. We say $M$ is *orientable* if an orientation exists, and otherwise we say $M$ is *nonorientable*. An *oriented manifold* is a pair $(M, \mathcal{O})$, where $\mathcal{O}$ is an orientation for $M$; an *oriented manifold with boundary* is defined similarly. The orientation for $T_pM$ is denoted by $\mathcal{O}_p$. We will often simply say "$M$ is an oriented manifold" and not explicitly name the chosen orientation.

Recall that a local frame is a collection of vector fields $\{E_i\}$ defined on some open $U \subseteq M$ which is linearly indepedent and spanning in the sense that each $\{E_i|_p\}$ is a basis for $T_pM$. Frames are automatically continuous.

**Exercise 4.2.2.** Let $(M, \mathcal{O})$ be an oriented $n$-manifold, $n \geq 1$. Show that any local frame with connected domain is either positively or negatively oriented. Show connectedness is necessary.

*Proof.* Let $\{E_i\}_{1\leq i\leq n} \subseteq \mathfrak{X}(U)$ be a local frame with $\pi_0(M) \cong *$.Let $s : U \to \{\pm 1\}$ be defined as follows: if $\{E_i|_p\}_{1\leq i\leq n}$ is positively oriented, then $s(p) = 1$; otherwise $s(p) = -1$. We must show $s$ is constant; since $U$ is connected it suffices to show $s$ is locally constant. We claim that since $\mathcal{O}$ is continuous, there exists $V \subseteq U$ open such that there is a local frame $\mathcal{F}$ on $V$ which is positively oriented. Indeed, we may simply choose a frame $\widetilde{\mathcal{F}}$ and restrict along $V$.

For each $x \in V$, let $A(x)$ be the transition matrix between the bases $\{E_i|_x\}_{1\leq i\leq n}$ and $\{F_i|_x\}_{1\leq i\leq n}$. Since $E$ and $\mathcal{F}$ are local continuous frames, we claim that $x \mapsto \det A(x)$ is continuous. Since $\det A(x) \neq 0$ for all $x$, this is a map $V \to \mathbb{R}\setminus \leq \{0\}$; since $V$ is connected we must have by continuity that the image of this map lies fully within a connected component of $\mathbb{R} \setminus \{0\}$. This immediately shows $s|V$ is constant, proving the first claim.

That $x \mapsto \det A(x)$ is continuous is trivial to show. The determinant is a polynomial in the entries of $A(x)$, and those can be recovered in a continuous manner from the entries of $\{E_i|_x\}$ and $\{\mathcal{F}_i|_x\}$ using continuity of the frames.

Now let $U \cong U_1 \amalg U_2$ be disconnected. I won't spell out the argument, but most of the argument is inducing a basis such that we can choose it to have one orientation on $U_1$ and the other on $U_2$. ■

**Proposition 4.2.3.** *Let $M$ be a smooth $n$-manifold. Any nonvanishing $n$-form $\omega \in \Omega^n(M)$ determines a unique orientation of $M$ for which $\omega_p$ is positively oritented for each $p$ in the sense given above: $\omega_p$ determines the orientation of $T_pM$ for all $p$. Conversely, if $(M, \mathcal{O})$ is an oriented manifold, there is a nonvanishing $n$-form that is positively oriented at each point, thus determining the orientation.*

*Proof.* Let $\omega \in \Omega^n(M)$ be nonvanishing. Then $\omega$ determines a pointwise orientation by proposition 4.1.5 since each $\omega_p$ is nonvanishing, so we need only show it is continuous. This is trivial for $n = 0$, so take $n \geq 1$. Let $p \in M$, and let $\{E_i\}_{1\leq i\leq n}$ be a local frame on some *connected* neighborhood $p \in U$. Let $\{\varepsilon^i\}_{1\leq i\leq n}$ be the dual coframe $\varepsilon^i|_p(E_j|_p) = \delta_{ij}$. On $U$, we may write[2] $\omega = f\varepsilon^1 \wedge \cdots \wedge \varepsilon^n$ for some continuous $f$. Since $\omega$ is nonvanishing, we have that $f$ is nonvanishing. . In particular, note that if $v = v^iE_i|_p \in T_pM$, then

$$\omega(v) = v^i\omega(E_i) = v^if(p)\left(\bigwedge_j \varepsilon^j\right)(E_i|_p) = v^if(p).$$

Take $v^i = 1$ such that $v = \sum_i E_i|_p$; then we recover $f(p) = \omega_p(E_1|_p, \ldots, E_n|_p)$. Thus $f = \omega \circ (\bigotimes_i E_i)$.

Since $f \neq 0$ for all $p \in U$, we have $f : U \to \mathbb{R} \setminus \{0\}$. Since $U$ is connected, it follows that $\operatorname{im} f$ lies fully within a connected component of $\mathbb{R} \setminus \{0\}$. Thus the frame is either positively or negatively

[2] On each point, $\omega_x = \omega_I^x \wedge \varepsilon^1 \cdots \wedge \varepsilon^n$ since the single vector is the basis, so define $f(x) = \omega_I^x$. Continuity is trivial; I think we can also assume smoothness but that's not needed here.

oriented. If negative, replace $E_1 \to -E_1$ to obtain a positively oriented frame by $n$-multilinearity. Since every point on a manifold has a connected neighborhood, this proves that the pointwise orientation determined by $\omega$ is continuous.

Conversely, let $(M, \mathcal{O})$ be an oriented manifold, and define $\Lambda^n_+ T^*M \subseteq \Lambda^n T^*M$ to be the set of all positively oriented $n$-covectors at all poins of $M$. Each intersection $\Lambda^n_+ T^*M \cap \Lambda^n T^*_p M$ is an open half-line since the vector space has dimension 1 and an orientation is a choice of connected component. There exists a partition of unity argument to construct a global smooth section of $\Lambda^n_+ T^*M$. ∎

In view of proposition 4.2.3, any nonvanishing $n$-form is often called an *orientation form*. If $M$ is oriented and $\omega$ induces the given orientation, we call it *positively oriented*; otherwise we call it *negatively oriented*. It can be checked that if $\omega$ and $\eta$ are both positively oriented, then $\omega = f\eta$ for some smooth strictly positive $f : M \to \mathbb{R}$. The existence of a function follows by proposition 4.1.5; smoothness follows by smoothness of $n$-forms.

**Terminology 4.2.4.** A chart $(U, \varphi)$ is said to be *positively oriented* if the frame $\{\partial_i\}$ is positively oriented, and similarly for negatively oriented. An atlas $\{(U_\alpha, \varphi_\alpha)\}_{\alpha \in A}$ is said to be *consistently oriented* if for all $\alpha, \beta \in A$, the transition map $\varphi_\beta \circ \varphi_\alpha^{-1}$ has Jacobian of positive determinant on all of $\varphi_\alpha(U_\alpha \cap U_\beta)$.

**Proposition 4.2.5.** *Let $M$ be a smooth $n$-manifold, $n \geq 1$. Given any consistently oriented atlas $\mathcal{A}$ on $M$, there is a unique orientation of $M$ with the property that each chart is positively oriented. Conversely, if $M$ is oriented and either $\partial M = \varnothing$ or $\dim M > 1$, then the collection of all oriented charts is a consistently oriented atlas for $M$.*

*Proof.* Let $M$ be a consistently oriented smooth atlas. Each chart determines an orientation by proposition 4.1.5 and definition, and since the jacobian has positive determinant they determine the same orientation on overlap. Thus they determine the same orientation globally. Conversely, let $\partial M = \varnothing$ or $\dim M > 1$. If any chart is negatively oriented, replace $x^1 \to -x^1$ to induce a positive orientation. This doesn't work when $\dim M = 1$ and $\partial M \neq 0$ since we take the final coordinate on a boundary to be nonnegative by convention. ∎

**Proposition 4.2.6.** *Let $\{M_i\}_{1 \leq i \leq k}$ be orientable manifolds. Then there is a unique orientation on $\prod_i M_i$ called the* product orientation *characterized by the following property: if for each $1 \leq i \leq k$, the $n$-form $\omega_i$ is an orientation form for $M_i$, then $\bigwedge_i \pi_i^* \omega_i$ is an orientation form for the product orientation.*

*Proof.* Here $\pi_i : \prod_j M_j \to M_i$ is the natural projection, and as usual the pullback determines a map $\pi_i^* : \Omega^n(M_i) \to \Omega^n(\prod_j M_j)$. Taking the wedge product yields a $\dim \prod_j M_j$-form

$$\bigwedge_{1 \leq i \leq k} \pi_i^*(\omega_i) \in \Omega^{\dim \prod_{1 \leq j \leq k} M_j} \left( \prod_{1 \leq j \leq k} M_j \right).$$

Nonvanishing is immediate, so by proposition 4.2.3 it determines an orientation. It now suffices to show any two different forms determine the same orientation. Let $\{\omega_i\}, \{\eta_i\}$ be two collections of orientation forms for $\{M_i\}$. Then for each $i$ and each $p \in M_i$, we have $\omega_i|_p, \eta_i|_p$ determine the same orientation for $T_p M_i$. Recall that $\prod_i T_{p^i} M_i \cong T_p \prod_i M_i$. From here the argument is fairly trivial. ∎

**Proposition 4.2.7.** *Let $M$ be a connected, orientable $n$-manifold. Then $M$ has exactly two orientations. If two orientations agree at a point, they are equal.*

*Proof.* It is clear at this point that every $T_pM$ has exactly two orientations, which amount to choices of connected components of $\Lambda^n T_p^*M \setminus \{0\}$, of which there are two. Choose an orientation Since $M$ is connected, for any nonvanishing $n$-form $\omega$ the map $s : M \to \mathbb{R}$ given by mapping $p \mapsto 1$ if $\omega_p$ is positively oriented, and $-1$ otherwise, must be constant using the same argument as before. In particular, $s$ raises to a map $s : \Omega^n(M) \to \{1, -1, 0\}$ mapping nonvanishing forms to their orientation, and vanishing forms to 0. It suffices to show those mapped to $-1$ all determine the same orientation. I don't want to show this its obvious. The second proposition follows immediately. ■

**Proposition 4.2.8.** *Let $M$ be an oriented $n$-manifold and $D \subseteq M$ a submanifold of codimension 0. Then the orientation of $M$ restricts to an orientation of $D$. If $\omega$ is an orientation form for $M$ and $i : D \hookrightarrow M$, then $i^*\omega$ is an orientation form for $D$.*

*Proof.* The first statement is immediate. For each $p \in D$, since $\operatorname{codim} D = 0$ there is an isomorphism $T_pD \cong T_pM$, and thus $D$ inherits a pointwise orientation from $M$. Continuity of the pointwise orientation is similarly immediate. Additionally for $v \in T_pD$, we have $\iota^*\omega(v) = \omega(v)$, so the other statement is similarly immediate. ■

Let $F : M \to N$ be a local diffeomorphism between oriented manifolds of positive dimension. We say $F$ is *orientation-preserving* if for each $p \in M$, the isomorphism $F_{*p}$ takes oriented bases of $T_pM$ to oriented bases of $T_{F(p)}N$. We say $F$ is *orientation-reversing* if it takes oriented bases to negatively oriented bases. If $M, N$ are 0-manifolds, then $F$ is preserving if $p$ and $F(p)$ have the same orientation, and orientation-reversing if they have the opposite orientation.

**Proposition 4.2.9.** *Let $F : M \to N$ be a local diffeomorphism between oriented manifolds. The following are equivalent.*

1. *$F$ is orientation preserving.*
2. *The Jacobian of $F$ with respect to any coordinates has positive determinant.*
3. *For any positively oriented orientation form $\omega$ on $N$, the form $F^*\omega$ is a positively oriented form for $M$.*

*Proof.* Let $p \in M$. Since $F$ is a local diffeomorphism, $F_{*p} : T_pM \to T_{F(p)}N$ is an isomorphism.

($1 \iff 3$): Let $\omega \in \Omega^n(N)$ be a positively oriented orientation form. Let $\{e_1, \dots, e_n\}$ be a basis for $T_pM$. By definition of the pullback:

$$(F^*\omega)_p(e_1, \dots, e_n) = \omega_{F(p)}(F_{*p}(e_1), \dots, dF_p(e_n)).$$

Suppose $F$ is orientation preserving. If $\{e_i\}$ is a positively oriented basis, then $\{F_{*p}(e_i)\}$ is a positively oriented basis for $T_{F(p)}N$. Since $\omega$ is positively oriented, $\omega_{F(p)}$ evaluates to a positive number on any positive basis. Thus $(F^*\omega)_p(e_i) > 0$, so $F^*\omega$ is positively oriented. Conversely, suppose $F^*\omega$ is positively oriented. Let $(e_i)$ be a positively oriented basis for $T_pM$. Then $(F^*\omega)_p(e_i) > 0$, which implies $\omega_{F(p)}(dF_p(e_i)) > 0$. Since $\omega$ is a positively oriented form, a basis evaluates to a positive number if and only if the basis itself is positively oriented. Thus $(dF_p(e_i))$ is positively oriented, so $F$ preserves orientation.

($1 \iff 2$): Let $(U, \varphi)$ and $(V, \psi)$ be positively oriented charts centered at $p$ and $F(p)$ respectively, with coordinate bases $\{\partial_{x^i}\}$ and $\{\partial_{y^j}\}$. The Jacobian matrix $J$ of $F$ is the transition matrix of the linear map $dF_p$ with respect to these bases:

$$dF_p\left(\frac{\partial}{\partial x^i}\right) = J_i^j \frac{\partial}{\partial y^j}.$$

Since $\{\partial_{x^i}\}$ is a positively oriented basis for $T_pM$, $F$ is orientation preserving if and only if $\{dF_p(\partial_{x^i})\}$ is a positively oriented basis for $T_{F(p)}N$. Since $\{\partial_{y^j}\}$ is already a positively oriented basis for $T_{F(p)}N$, the basis $\{dF_p(\partial_{x^i})\}$ is consistent with $\{\partial_{y^j}\}$ (i.e., positively oriented) if and only if the determinant of the transition matrix $J$ is positive. ∎

**Remark 4.2.10.** Composites of orientation-preserving maps are orientation-preserving. This follows immediately by either the definition, or proposition 4.2.9 (2).

**Proposition 4.2.11.** *Let $F : M \to N$ be a local diffeomorphism and $N$ oriented. Then there is a unique orientation on $M$ such that $F$ is orientation preserving. We call this orientation the* pullback orientation *induced by $F$.*

*Proof.* For each $p \in M$, there is a unique orientation on $T_pM$ that makes the isomorphism $F_{*p} : T_pM \cong T_{F(p)}N$ orientation preserving defined in the obvious way. This proves uniqueness of the pointwise orientaiton. To show this pointwise orientation is continuous, let $\omega$ be a smooth orientation form for $N$, and note that $F^*\omega$ is a smooth orientation form for $M$ by proposition 4.2.9. This defines an orientation which coencides with the pointwise orientation. ∎

**Notation 4.2.12.** If $(N, \mathcal{O})$ is an oriented manifold and $F : M \to N$ is a local diffeomorphism, the pullback orientation on $M$ induced by $F$ is often denoted $F^*\mathcal{O}$.

**Proposition 4.2.13.** *The pullback orientation is contravariant.*

*Proof.* Let $F : M \to N$ and $G : N \to P$ be local diffeomorphisms and $(P, \mathcal{O})$ an oriented manifold. We must show $(G \circ F)^*\mathcal{O} = F^*G^*\mathcal{O}$. Let $\omega$ be an orientation form for $P$. Then $(G \circ F)^* = F^*G^*\omega$ is an orientation form for $M$ by proposition 4.2.9. ∎

## 4.3 Boundary orientations

An (embedded, immersed) hypersurface $S$ is an (embedded, immersed) submanifold $S \subseteq M$ of codimension 1. A vector field *along* $S$ is a map $S \to TM$ which is a section of $\pi : TM \to M$ in the relevant sense.

**Proposition 4.3.1.** *Let $M$ be an orientable $n$-manifold, $S$ an immersed hypersurface, and $N$ a vector field along $S$ that is nowhere tangent[3] to $S$. Then $S$ has a unique orientation such that for each $p \in S$, $\{e_i\}_{1\leq i\leq n-1}$ is an oriented basis for $T_pS$ if and only if $\{N_p\} \cup \{e_i\}_{1\leq i\leq n-1}$ is an oriented basis for $T_pM$. If $\omega$ is an orientation form for $M$, then $i^*(\iota_N\omega) = i^*\omega(N, -)$ is an orientation form for $S$ with respect to this orientation, where $i : S \hookrightarrow M$.*

*Proof.* Let $\omega$ be an orientation form for $M$. Then $\sigma = i^*(\iota_N\omega)$ is an $(n-1)$-form on $S$, since the pullback $i^*$ is really just the restriction of the interior product $\iota_N\omega$ to $S$. It follows taht $\sigma$ is an orientation if and only if it never vanishes. Since $N$ is nowhere tangent to $S$ and $S$ has codimension 1, for any basis $\{e_i\}_{1\leq i\leq n-1}$ for $T_pS$, the set $\{N_p, e_1, \dots, e_{n-1}\}$ is a basis for $T_pM$ since $N_p$ must be independent of $\{e_i\}_{1\leq i\leq n}$. Since $\omega$ is nowhere vanishing,

$$\sigma(e_1, \dots, e_{n-1}) = \omega(N_p, e_1, \dots, e_{n-1}) \neq 0.$$

Thus $\sigma$ is nowhere vanishing. The orientation is clearly the same as the one in the proposition. ∎

[3]This means that $N_p \in T_pM \setminus T_pS$.

**Corollary 4.3.2.** *Let $M$ be an orientable manifold with boundary. Then $\partial M$ is orientable. In particular, any vector field outward pointing*[4] *along $\partial M$ determines the same orientation.*

I decline to prove the second part of corollary 4.3.2.

[4] Along and the final component in coordinates is negative.

# Chapter 5

# Integration on manifolds

**Lecture 30: Personal Review**

Tue 13 Jan 2026 22:09

## 5.1 Review of some notions

**Definition 5.1.1.** Let $F : M \to \mathbb{R}$ be a function. The *support* $\operatorname{supp} F$ is the closure of the set of all points wherein $F$ is nonzero

$$\operatorname{supp} F \coloneqq \overline{\{x \in M \mid F(x) \neq 0\}}.$$

Given $U \subseteq M$, we say $F$ is *supported* in $U$ if $\operatorname{supp} F \subseteq U$. If $\operatorname{supp} F$ is compact, we say $F$ is *compactly supported*.

**Definition 5.1.2.** Let $A \subseteq M$ be a closed set. A *smooth bump function* for $A$ is a function $\psi : M \to [0, 1]$ which is identically 1 on $A$, and there exists an open neighborhood $A \subseteq U \subseteq M$ wherein $0 \leq \psi|U \leq 1$, and $\psi = 0$ outside of $U$. We often specify the neighborhood $U$, leading to the following alternate definition. A smooth bump function for $A$ supported in $U$ is a function $\psi : M \to [0, 1]$ supported in $U$ with $\psi|A$ the constant function at 1.

**Definition 5.1.3.** Let $\mathcal{X} = \{X_\alpha\}_{\alpha\in A}$ be an open cover of $M$. A *partition of unity subordinate to* $\mathcal{X}$ is a collection of continuous functions $\{\psi_\alpha : X_\alpha \to [0, 1]\}_{\alpha\in A}$ such that the following properties are satisfied:

1. $\psi_\alpha$ is supported in $X_\alpha$ for each $\alpha \in A$;
2. For all $p \in M$, there is a neighborhood $U$ of $p$ which intersects only finitely many $\operatorname{supp} \psi_\alpha$. We say the family of supports is *locally finite*;
3. For all $p \in M$,

   $$\sum_{\alpha\in A} \psi_\alpha(p) = 1.$$

   Note that the local finiteness condition in particular implies that $\psi_\alpha(p)$ is nonzero for only finitely many $\alpha$, so there are no convergence issues.

If each $\psi_\alpha$ is smooth, we say that partition of unity is *smooth*.

Some theorems.

1. Let $\mathcal{X} = \{X_\alpha\}_{\alpha \in A}$ be an open cover of $M$. Then there exists a smooth partition of unity subordinate to $\mathcal{X}$.

2. Given any closed set $A$ and any open neighborhood $U$ of $A$, there is a smooth bump function for $A$ supported in $U$.

3. Given a smooth map $f : A \to \mathbb{R}^k$ with $A$ closed, for all open $U \subseteq M$ containing $A$ there exists an extension $\widetilde{f} : M \to \mathbb{R}^k$ with $\widetilde{f}|A = f$ and $\operatorname{supp} f \subseteq U$. In particular, $f$ vanishes outside of $U$.

4. Let $M$ be a manifold. If $K$ is a closed subset, then there is a smooth nonnegative function $f : M \to \mathbb{R}$ with $f^{-1}(0) = K$.

**Remark 5.1.4.** Functions vanish outside of their support. The main takeaway from saying a function is supported in a set is that it vanishes outside of that set.

## 5.2 Integration of differential forms

**Definition 5.2.1.** A *domain of integration* is a bounded subset $D \subseteq \mathbb{R}^n$ whose boundary has measure zero.

For a general $n$-form on $\mathbb{R}^n$, our form must be of the form $\omega = f \mathrm{d}x^1 \wedge \ldots \wedge \mathrm{d}x^n$ for some continuous $f$. We can define the integral of it over some domain of integration $D \subseteq \mathbb{R}^n$ as

$$\int \omega = \int_D f \, \mathrm{d}V = \int_D f \, \mathrm{d}x^1 \cdots \mathrm{d}x^n.$$

We can generalize slightly more. Define the *support* of an $n$-form $\omega \in \Omega^n(M)$ as the closure

$$\operatorname{supp} \omega = \overline{\{p \in M \mid \omega_p \neq 0\}}.$$

Returning to $M = \mathbb{R}^n$ to build intuition, suppose that $\omega$ is compactly supported on some open $U \subseteq \mathbb{R}^n$. We can then define

$$\int_U \omega := \int_D \omega$$

for any domain of integration $D$ containing $\operatorname{supp} \omega$. It is immediate that this definition is independent of the choice of $D$.

# Chapter 6

# De Rham cohomology

**Lecture 31: Personal Review**

Thu 15 Jan 2026 22:19

## 6.1 The de Rham cohomology groups

Recall that a closed form is a form $\omega$ such that $\mathrm{d}\omega = 0$, and an exact form is any form of the form $\mathrm{d}\eta$. Since $\mathrm{d}^2 = 0$, clearly exact implies closed. All closed forms are *locally* exact, but not necessarily globally exact, meaning that the question of whether closed implies exact depends entirely on the global topological properties of the manifold.

Since $\mathrm{d}^2 = 0$, defining $\Omega^k(M) = 0$ for all $k < 0$ or $k > n$, then we have a cochain complex

$$\Omega^0(M) \xrightarrow{\mathrm{d}} \Omega^1(M) \xrightarrow{\mathrm{d}} \Omega^2(M) \xrightarrow{\mathrm{d}} \cdots$$

with values in $\mathbb{R}$-vector spaces. The $k$th cohomology group can then be described as the quotient of the set of closed $k$-forms by the set of exact $k$-forms, and thus closed implies exact if and only if the cochain complex has vanishsing homology in degree $k \geq 1$. The cohomology groups are called the *de Rham cohomology groups*, and are denoted $H^k_{\mathrm{dR}}(M)$.

**Lemma 6.1.1.** *Let $F : M \to N$ be smooth. Since there exist maps $F^* : \Omega^k(N) \to \Omega^k(M)$ for all $N$, there exists a functor $\mathcal{M}\mathrm{an}^{\mathrm{op}} \to \mathcal{C}\mathrm{h}(\mathbb{R})$ given by mapping manifolds to their de Rham cochain complexes, and by mapping $F$ to the chain map given by pullbacks.*

*Proof.* Since pullbacks are functorial, it suffices to show naturality, meaning $F^* \circ \mathrm{d} = \mathrm{d} \circ F^*$. This follows by 3.1.22 (4). ∎

**Corollary 6.1.2.** *The map $M \mapsto H^k_{\mathrm{dR}}(M)$ is functorial $\mathcal{M}\mathrm{an}^{\mathrm{op}} \to \mathbb{R}\text{-}\mathcal{V}\mathrm{ect}$.*

*Proof.* Homology is functorial. In particular, any smooth $F : M \to N$ lowers to a map $F^* : H^k_{\mathrm{dR}}(N) \to H^k_{\mathrm{dR}}(M)$ defined by $F^*[\omega] = [F^*\omega]$. ∎

**Corollary 6.1.3.** *Diffeomorphic manifolds have isomorphic de Rham cohomology groups. Even more powerfully, diffeomorphic manifolds have isomorphic (not just quasi-isomorphic) de Rham complexes of differential graded algebras.*

## 6.2 Elementary computations

**Proposition 6.2.1.** *Let $\{M_j\}_{j\in J}$ be a countable collection of smooth manifolds. Then $H^k_{\mathrm{dR}}$ preserves countable products in the sense that*

$$\prod_{j\in J} H^k_{\mathrm{dR}}(M) \cong H^k_{\mathrm{dR}} \coprod_{j\in J} M_j.$$

*Proof.* It suffices to show $M \mapsto \Omega^k$ preserves products in this sense, since limits in functor categories are computed pointwise, $\mathcal{C}\mathrm{h}(\mathbb{R})$ can be realized as a functor category, and homology preserves countable products. The isomorphism for $\Omega^k$ is given by the inclusion maps $i_j : M_j \hookrightarrow \coprod_j M_j$ by

$$\Omega^k \coprod_{j\in J} M_j \ni \omega \mapsto \left\{i_j^* \omega\right\}_{j\in J} \in \prod_{j\in J} \Omega^k(M_j).$$

I find this obvious and omit the proof. ■

**Proposition 6.2.2.** *Let $M$ be a connected smooth manifold. Then $H^0_{\mathrm{dR}}(M)$ is the space of constant functions, and thus 1-dimensional.*

*Proof.* There are no $(-1)$-forms, and thus there are no exact 0-forms. A 0-form is a smooth function $f : M \to \mathbb{R}$, and a closed 0-form is a function $f$ such that $\mathrm{d}f = 0$. Recall this is true if and only if $f$ is constant. ■

**Proposition 6.2.3.** *Let $M$ be a 0-manifold. Then the de Rham cohomology of $M$ is concentrated in degree 0, and is a direct product of 1-dimensional vector spaces for each point of $M$.*

## 6.3 Homotopy invariance

There is a strengthening of corollary 6.1.3.